

2 Literature Review

2.1. Introduction and methodology adopted

The selection of the articles utilized in this literature review was carried out following an approach inspired by the PRISMA framework. Specifically, the process was structured into three main phases.

The first phase was the research and identification of a specific Scopus search string. Being the topic of the research quite specific, it was necessary to build a string which will encompass all the different nuances and meanings; after few attempts the final string was the following:

(TITLE-ABS-KEY ((economic AND benefit*) OR revenue* OR incentives OR profit* OR introit* OR incom*) AND TITLE-ABS-KEY (recs OR (energy PRE/0 communit*)) AND TITLE-ABS-KEY ((redistribut* OR reallocat* OR sharing OR distribut* OR allocat*)))

An immediate constraint was placed on the string, including only papers published after the 2018 RED II directive, in order to consider studies concerning energy communities that comply with the characteristics coming from the European directive. This resulted in an initial sample of 497 papers.

Starting from this sample, the second phase of paper screening began, by reading the titles, abstracts and keywords. Specific exclusion criteria were analysed, such as not focusing enough on RECs or not going deep enough into the issue of redistribution of benefits. 47 articles were the result of this phase, and identified as potentially relevant.

The third phase was the reading of the papers, with special focus on the methodologies used and the conclusions. Of the 47 papers, 28 were cited and included in the literature review.

This overall process was designed to ensure rigor and transparency in the selection of the literature.

The literature shows an increased level of heterogeneity in nature and amount of the benefits coming from management of Renewable Energy Communities, varying significantly from one national context to another, reflecting the different support policies and regulatory frameworks adopted in EU countries. In some countries, such as Italy, incentives focus on energy shared virtually between members (virtual net metering), while in other contexts - e.g. in Portugal or in some cooperative models in

Germany - there is also the valorisation of energy fed into the grid, with compensation mechanisms for energy sold locally. For instance, in Germany, as expressed by Minuto & Lanzini [3] electricity fed into the grid is compensated by a special feed-in tariff guaranteed by the government for 20 years.

In parallel, Local Energy Markets (LEMs) are spreading, markets within energy communities where price and quantity of energy exchanged between peers are set according to local and dynamic criteria.

In this heterogeneous scenario, the issue of redistribution of economic benefits among the members of a REC assumes a central role, not only for the financial equilibrium of the community, but also to ensure its equity, cohesion and long-term sustainability. The scientific literature has proposed several approaches to address this issue, ranging from mathematical models based on game theory to simpler methods based on criteria of consumption, investment or socioeconomic vulnerability.

It must be stressed, however, that the aforementioned simplified models mainly refer to the Italian context, where the allocation of shared energy takes place in a centralised and “a posteriori” manner, through GSE flow management.

Mathematical models based on game theory, on the other hand, also analyse REC models with dynamic allocation, such as P2P or the specific case of LEMs, adopted at European level.

Finally, a brief overview on some “European” models will be provided, commenting some dynamic pricing mechanisms, with allocation of energy shared made “a priori” and enhancing user engagement and active participation.

This literature review aims to classify and critically discuss the main benefit redistribution methods in RECs, while identifying factors influencing their adoption, open challenges and remaining gaps in the scientific landscape.

2.2. Methodologies classification

2.2.1. Mathematical models based on game theory

Great part of literature regarding redistribution schemes addresses the topic adopting various types of mathematical models based on game theory.

From analysing some papers in literature ([4], [5], [6] [7]) a comprehensive explanation of what game theory is and how it is a valuable tool for allocating economic benefits in RECs is followingly proposed.

Game theory is an effective tool to address the interactive nature of energy sharing, since it provides general mathematical techniques for analysing situations in which two or more individuals make decisions that will influence one another’s welfare.

The game describes any situation in which the decision-making processes of the players are interrelated. The players are said to be rational, as their decisions are guided solely by the desire to complete their own objectives, that are assumed to be the maximization of their expected payoff, mathematically described by an utility function.

This model can be used to simulate the behaviour of a REC, with the “players” being in this case the various constituent members. The economical value generated hour by hour depends on the presence and the interaction between load and production, in particular on the quantities of energy produced and shared. The players can decide to take part to the community or not. Both producers and consumers want to take advantage from the participation to the community. Producers can be considered as external actors that can invest in the installation of new generators and wants to get a return from the investment, or as community members who invest in a plant, consume what they need themselves, and make the excess energy available to other users so that it can be shared.

Game-theoretic models for energy communities require a precise modelling of energy profiles, sharing efficiency, and economic outcomes. This means calculating, for each possible coalition of members, the actual savings, incentives, and benefits they would generate by cooperating, based on consumption/production patterns, regulatory frameworks, and technical constraints.

Two approaches are identified in the theory, according to the fact that players are or not collaborating for achieving maximum social welfare.

In particular, in *Non-cooperative approaches* each user egoistically operates its system to maximize its benefits, and may compete with other users for the same resources.

This is, for instance, the case of Local Electricity Markets, in which users are able to trade in a competitive environment.

Cooperative games are instead characterized by the possibility of communication between players, who can form coalitions between each other in order to improve their payoff from the game. The coalition shall be stable, i.e. the total reward shall be fairly distributed among users in order to adhere to two main principles: rationality and efficiency.

This model suits better with the case of REC since users are not typically in direct trading and would rather delegate this role to others, and considering furthermore the social focus of EC policies expressed by EU.

The following describes the main approaches by game theory that can be used for redistributing economic benefits.

Without accurate input data, the results of cooperative game theory (like the Core, Shapley, or Nucleolus) are not reliable or actionable.

Core

The “Core” allocation mechanism represents the set of the possible stable reward distributions that satisfy the previously mentioned principles of *rationality* and *efficiency*.

A profit distribution for all users of a given (grand) coalition S is rational when any sub-coalition is awarded a higher profit in the grand coalition N (e.g., the energy community) than in a separated coalition on their own.

A distribution is efficient when the collective benefits are completely distributed among all players.

The “core” delineates a space of solutions x that can guarantee these properties.

In mathematical terms, a payoff vector x is in the core of a coalitional game $(N; v(s))$ if and only if:

$$\forall S \subseteq N, \sum x_i \geq v(S) \quad (2)$$

With $v(S)$ being the “value” function of any possible coalition S of players within the grand coalition.

Shapley

All thing considered, a solution based on the Core may not reflect the fairest way to allocate the grand’s coalition value.

Going beyond this limit, the Shapley Value takes into account the added value each player brings to the coalition, reflecting the fairest payoff for the players in the large coalition.

The Shapley value, basis to allocate economic benefits to the members, is calculated as follows:

$$\varphi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} (v(S) - v(S \setminus \{i\})) \quad (3)$$

Where $|S|$ refers to the size of the coalition S and $(v(S) - v(S \setminus \{i\}))$ is the contribution of the user i to the overall value of the coalition S , expressing the concept of “Marginal Contribution”.

The final value assigned to each user will represent his weighted average marginal contribution to the value of any coalition, within the grand coalition.

From the Shapley calculation the payoff related to each user can be obtained, that relates to his average contributions to the daily operational profit (or cost) in all the different coalitions.

The Shapley value can be considered a fair allocation mechanism in the sense that prosumers, who promote self-consumption and virtual energy sharing within the REC, receive higher payoffs compared to consumers (in any case the latter have economic convenience to be part of the REC instead of operating independently).

However, one of the main drawbacks of the Shapley value consist in its computational burden as the number of players inside the coalition increases. In particular, since it requires to consider 2^N coalitions, it will be computationally demanding to evaluate the Shapley Value for a set of hundreds of users like it could be the one of an energy community.

Nucleolus

The Nucleolus approach is based on an iterative optimization procedure aiming at minimizing the discontent of users in the community. The procedure focuses on minimizing the “dissatisfaction” of the members of each coalitions, if their benefits allocated within the grand coalition (final energy community) are lower than those they could obtain in independent coalitions outside of the grand coalition. The aim of the algorithm is to progressively minimize the unhappiness of the most unhappy coalition in every iteration.

It can be proven, as from the paper by Volpato et. Al [8], that this approach leads to a unique solution when a solution exists and guarantees the concept of stability (according to “collective fairness”), rewarding members whose benefits allocated within the grand coalition are lower than those they could obtain in independent coalitions outside of the grand coalition.

However, similarly to the Shapley value, it involves the solution of a number of optimization problems that grows exponentially with the size of the community. Moreover, the Nucleolus does not necessarily satisfy the fairness axioms postulated by Shapley since higher contributions of some users to the benefits of the EC and its coalitions could be less rewarded. Therefore it is arguable whether it provides a fair distribution of profits.

Nash Equilibrium

The Nash Bargaining (NB) optimization approach allows for the simultaneous optimization of the total benefit of the EC and the individual benefits of its members compared to a “disagreement” case in which each member acts individually (i.e., without cooperation).

The objective function to be maximized is the Nash product:

$$\prod_i u_i^{incr} \quad (4)$$

With u_i^{incr} being the incremental economic benefits of users derived from their cooperation within the EC compared to the “disagreement” case.

According to cooperative game theory, the NB solution isn’t just efficient and individually rational (as Core approach is) but also is Pareto optimal (i.e., a user cannot increase its optimal incremental benefit without decreasing the incremental benefits of other users).

Furthermore, the main advantage of this approach (as the Shapley one) over other methods like the core, is the existence and uniqueness of the solution.

However, the main issue with the NB is that the objective function in the Equation is non-convex and non-linear, which complicates the search for the global optimal solution for both the system and the users.

2.2.2. Approximation of Shapley Value – VCG mechanisms

The use of models based on game theory, and in particular on Shapley Value, while attracting interest in the subject of redistribution of economic benefits, still presents a computational challenge, especially if we refer to RECs with more than a dozen users. Much of the literature ([9], [10], [11], [12]) has tried to find alternative methods with which to approximate its value, while attempting to reduce its computational complexity.

Many articles have identified a mechanism belonging to auction theory that attempts to reflect the meaning belonging to Shapley Value: the Vickrey-Clarke-Groves (VCG). Underlying this mechanism is the identification of the so-called marginal contribution (MC), being the difference in financial savings (*sav*), looking at the reduction of the overall bills summed, between the base case and the case without that user.

$$MC_i = sav(EC) - sav(EC \setminus \{i\}) \quad (5)$$

As shown in the formula cited in [10], the savings here refer to overall economic benefit to the community measured in terms of lower energy costs and incentives received by shared energy valorised.

After finding all the contributions, it is possible to determine the benefit distribution vector, by subdividing total economic benefits (TEB) between users weighting by their respective contribution.

$$BEN_i = TEB * \frac{MC_i}{\sum_i MC_i} \quad (6)$$

The marginal contribution approach proposed by VCG uses the same intuition as in Shapley value calculation, but, whereas the Shapley value takes the mean marginal

contribution of agent i for every possible subcoalition in the community, this method computes the benefits redistribution by only looking at the last marginal contribution, making it a much more time-efficient method.

This method should ensure that the benefit percentage assigned to a user does not trigger discontent among other members, who would not have equally access to that amount without

that member participation. According to the community composition, the distribution vector may also have null terms. We can have a null term if a member is not providing any benefit.

However, when applicable to the real case, in necessary cases the REC could proceed according to other criteria and might still want to allocate something to this member (i.e., in energy poverty, where people are unable to access energy).

2.2.3. Simplified Approaches

Although models based on game theory (such as the Shapley value) and auction theory (such as the Vickrey-Clarke-Groves mechanism) provide theoretically robust tools to ensure fairness and efficiency in the redistribution of economic benefits within Renewable Energy Communities (RECs), their practical application is often complex. These approaches require a large amount of information, detailed analysis of participants' energy profiles, and multiple simulations to evaluate alternative consumption and production scenarios.

For these reasons, recent literature has proposed and analyzed simplified allocation models, which, while sacrificing some theoretical precision, offer greater transparency, scalability, and ease of implementation. These models are typically based on more intuitive rules, such as the share of energy contributed or consumed, actual participation in collective self-consumption, or simple proportional allocation criteria.

Equal Share Allocation Method

This method distributes total benefits (TEB) in equal parts to the members, where n is the number of EC members:

$$BEN_i = \frac{TEB}{n} \quad (7)$$

Without taking into account the contribution to the overall demand or generation, or the social purpose of the community, it distributes the value evenly. It is a really straightforward approach that is simple to comprehend and use, but it offers no motivation to lessen load or boost self-generation. The method's fairness is dubious since each person's input isn't taken into account.

Energy Sharing based

These methodologies approach the issue of distribution from a point of view of shared energy contribution by each member within the REC.

Numerous models in the literature have based the redistribution of economic benefits in the REC on the contribution of each member to the shared energy.

Fioriti et al. [6] propose a sharing energy scheme that uses two equivalent reward tariffs for producers and consumers: π_{pr} which is the tariff related to shared produced energy and π_{con} which is the tariff related to shared consumed energy.

Being E^{pr} and E^{con} the shared produced energy and the shared consumed energy respectively, the reward to each member is calculated as follows:

$$BEN_i = \pi_{pr} * E^{pr} + \pi_{con} * E^{con} \quad (8)$$

It is important to say that in this paper the authors assume the two tariffs to be equal, giving same importance to produced and consumed shared energy.

Trevisan et al. [13] proposes a method of allocating economic benefits that distinguishes according to whether the user is a producer or a consumer. The benefits to be allocated to producers will be divided according to the producer's proportional contribution to the energy input in a given hourly frame. Similarly for consumers, the load of each will be looked at to allocate the individual share proportionally.

Although it explores how to allocate economic benefits within the same category of users (consumers or producers), it does not introduce any distinction between the allocated benefits to be assigned to producers and the amount to be assigned to consumers.

In the studies by Campagna et al. [14] and Minuto et al. [3], the redistribution of benefits is based on the energy shared, looking at the different loads and demands of the users.

In the first, the quota assigned to each consumer is given as a proportion to the overall load of the energy community, considering that specific time frame (typically, 1 hour).

Minuto et al. propose instead an algorithm to allocate energy in order not to disadvantage those who consume less (e.g. private consumers versus businesses). The basic idea is to allocate a fixed amount of shared energy (and therefore incentive) to each user, and then distribute the remainder based on the contribution made by the user in terms of shared energy.

Volpato et al. [15] uses a method called 'Uniform Pricing', based on the identification of a 'price' calculated as the ratio between the total benefit derived from the REC and the total demand for the REC.

The users' contribution will be simply calculated by multiplying the Uniform Price by the energy demand of each user.

While these three last methods are useful for allocating benefits among consumers by looking at each one's load, they do not take into account the allocation of shared energy input.

Power Based

These methods redistribute the benefits of the REC among members according to the installed power of the plants by each member.

This can be done directly allocating benefits proportionally to the total power owned by each member, such as in the paper by Bâra et al [16], or by separating the total benefits between consumers and producers, looking at the power of the plants or the meters' power, by using some specific coefficients [17].

Ownership/financing based

These mechanisms act in such a way as to protect those who have invested their resources to acquire the REC production facilities. Various mechanisms for financing the plants exist, as reported in the paper by Minuto et al. [3]. Individual REC members may agree to split the amount to purchase the plants in the REC equally, they may resort to an external request such as a bank transfer, or they may contact a third-party supplier to finance the transaction. Furthermore, often in RECs the plants are purchased by the individual member who has ownership of that plant and who decides to share the surplus energy produced.

Benefits allocated to each member are then allocated following this formula:

$$BEN_i = TEB * Ow_i \quad (9)$$

Being Ow_i the ownership share of community's power plants by user i.

If, for instance, plants are financed through an equal support by each member, then the formula can be also seen as:

$$BEN_i = \frac{TEB}{n} \quad (10)$$

Fioriti et al. [6] add to this a specific fixed quota for each member, regardless of whether the user financed the installation or not.

In the paper by Fina [18] 'Contribution' is manifested in form of assets that are essential for the operation of the EC, e.g. PV systems.

To reward such contribution, EC-incurred grid costs of participants with a PV system are redistributed to be borne by community members without a PV system.

Finally, a third-party investment feasibility according to different cost allocation methods (through dynamic pricing), is analyzed in the paper by Li et al [19], showing that is economically feasible for a third party to invest in an energy community with the right energy prices and payback time.

Multicriteria Approaches

Several times within the literature, a multi-criteria approach has been proposed to address the issue of redistribution of economic benefits in RECs.

These methodologies attempt to combine the different perspectives we analysed earlier in the review, such as the one that benefited those who financed the plants, rather than those who contributed the most to the shared energy, in order to ensure a redistribution that takes into account various important aspects within the mechanisms of RECs.

It is common in the literature to divide the process into two stages: in the first the total amount of benefits is divided between the different categories of members (consumers, producers and prosumers), in the second the benefits are allocated to the individual members looking at their contribution to energy production and consumption.

Such an approach is used in the paper by Trevisan et al. [20].

They use a tool called the “Benefits Redistribution Tree”, within which the total amount derived from state incentives is divided by various allocations into specific “shares”.

As can be seen from Figure 5, each category will be allocated a share of the incentive, taking into account a specific constraint, and according to a specific optimization function to be optimised.

The constraint has the objective to find the ideal apportionment such that the incentive received by every user plus the corresponding RID, is greater than a certain percentage of the costs it incurs.

$$\sum_{l \in L} a_{il} q_l + r_i \geq b_i z_i, \quad \forall i \in I \quad (11)$$

Where a_{il} is the element in position (i,l) of the matrix A, showing the contribution of each member to the energy withdrawn from or injected into the grid, and identifies the percentage of participation in the l-th category by the i-th user; q_l is the share at the l-th leaf node, whilst r_i , b_i and z_i are respectively the RID income, the incurred costs that are to be lowered and the percentage of minimum desired reduction for the i-th user.

The optimization function can vary, depending on the different objectives, such as maximizing the incentive redistributed, minimising it (but maintaining a constraint on the covering of costs), or maximizing the cost reduction for producer and prosumers.

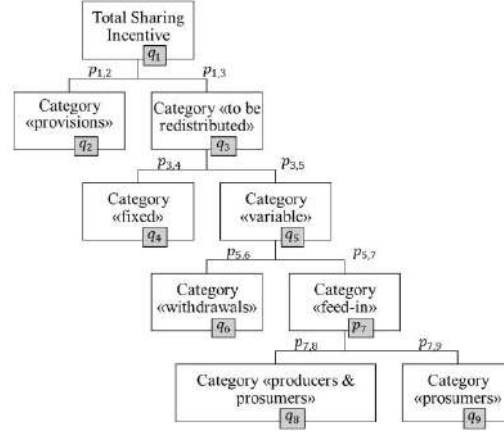


Figure 5 : Economics benefits redistribution tree proposed by Trevisan

Similar approaches are adopted in the paper by Minuto et al. [3] and Candela et al. [17].

Specifically, in the second paper the total benefits are first subdivided into two main categories utilizing two coefficients, alpha and beta, respectively representing the percentage of incentives to be redistributed to the consumers and the percentage to be redistributed to producers.

It must be emphasised that the proposed approach uses the values of total capacity available at consumers and producers to identify the value of the two coefficients.

Prosumers will get benefits from both the allocation to the producers and to consumers, according to their quota.

Following the formula to calculate economic benefits to consumers and to producers.

As we said before prosumers will get both portions, according to their contribution.

$$BEN_i = \alpha * TEB * \frac{E_i^{con}}{\sum_{i \in C} E_i^{con}} \quad (12)$$

$$BEN_y = \beta * TEB * \frac{E_y^{pr}}{\sum_{y \in P} E_y^{pr}} \quad (13)$$

Specifically, i represents the generic consumer belonging to class C (consumers); y represents the generic producer belonging to class P (producers).

It is necessary to point out that these values refer just to the hourly energy consumed and produced, but not looking at the energy shared.

For this reason, a second version of the method is suggested, in which a constraint is introduced. This refers to the introduction of a minimum threshold for energy consumed by a passive user, with respect to the total amount of shared energy. If the energy consumed by the generic passive user is greater than the threshold, then he will receive the incentives, otherwise he will receive nothing.

The paper by Romagnosi et al. [21] adopt a different approach, including priorities for fragile families while redistributing benefits among members.

More specifically, two scenarios are suggested: in the first one the distribution is based on the contribution to the sharing, guaranteeing priority to fragile families; in the second one, instead, is suggested a different distribution for prosumer, based on their PV system ownership percentage, and for final consumers, based on the contribution to the sharing, guaranteeing priority to

fragile families.

Following this last approach, in the paper by Campagna et al. [14] is proposed a distribution mechanism that try to consider those three different allocation methods simultaneously.

The approach is very close to the one already analysed in the section with methods “Based on energy poverty and social inclusion”, where the $n\%$ rule is adopted to define $share_{EP}$.

The main difference is that in this “tri-level” method the maximum share is 33%.

The remainder is split using “Ownership”-based method (that redistributes the selling income) and the “Proportional” one (that redistributes the sharing income).

Multicriteria approaches with focus on Energy Poverty and Social inclusion

A significant part of the literature deals with energy poverty as one of the topics underlying the emergence and development of renewable energy communities. energy poverty is

defined by the European Commission (EC) as a situation where “energy bills represent a high percentage of consumers’ income, or when they must reduce their household’s energy consumption to a degree that negatively impacts their health and well-being”.

A valid measure to assess energy poverty is required and proposed by Campagna et al. [14]. They introduce a set of three indicators included in the EU Survey related to Income and Living Conditions. These indicators are the inability to keep the house warm, the arrears on the bills, and the presence of leaks or dampness in the building. The subsistence of one or more of the mentioned conditions is generally perceived as evidence of energy poverty risk.

Related to this, the paper by Hanke et al. [22] included the notion of energy justice with respect to the potential given by RECs to enable citizen participation in energy transition and shape a just transition.

Drawing on data collected among 71 European RECs, their research investigated how RECs engage in this social role, specifically addressing how low-income and energy-poor households could benefit from granting access to affordable energy tariffs and energy efficiency measures when participating in RECs.

Referring to distribution of benefits in the REC, in the paper by Campagna et al. [14] is described a method that attributes a share ($n\%$) of REC revenues to each user in risk of energy poverty based on LIHC.

The LIHC (Low Income High Costs) indicator here categorizes households at risk of energy poverty if they simultaneously have an energy expenditure above the national median value $P50(s_{e,i})$ and family income net of energy expenditure ($y_i - s_{e,i}$), divided by the income recipients in the household (N_{inc}) below a threshold value (y^*).

$$LIHC = I\{[s_{e,i} > P50(s_{e,i})] \cup [\frac{y_i - s_{e,i}}{N_{inc}} < y^*]\} \quad (14)$$

Therefore, the total share of revenues for energy poverty mitigation ($share_{EP}$) is $n\%$ times the total number of vulnerable users in the REC (N_{vu}) up to a maximum of 50% as share, as follows:

$$share_{EP} = \min(50\%, n\% \cdot N_{vu}) \quad (15)$$

It is also important to mention the more advanced versions of LIHC provided by [23].

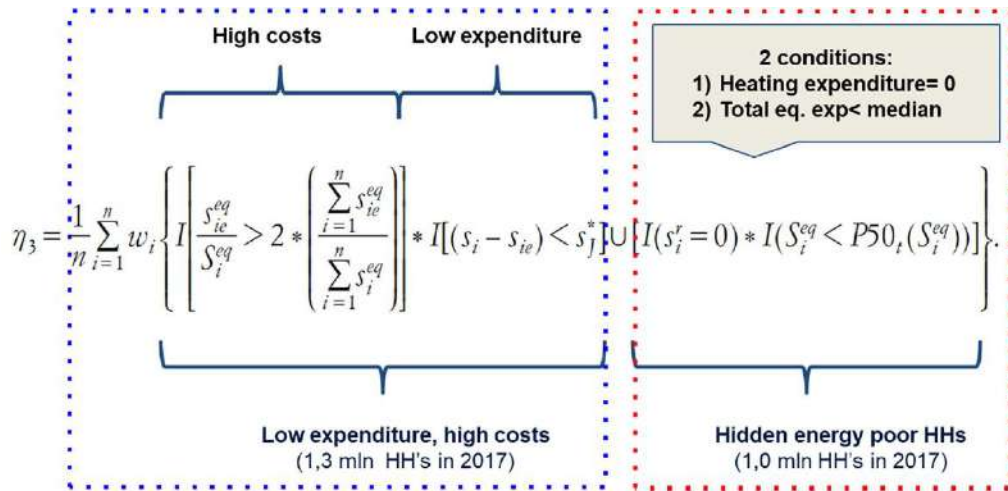


Figure 6: Advanced version of LIHC indicator

In this paper the LIHC indicator is partially modified: to identify a family with high costs, its energy expenditure is compared to its total expenditure and it is verified that this ratio is more than double the national average. To identify “low income” families, expenditures are used as a proxy of income, and then compared with the relative

poverty threshold (net of energy costs). A second part refers instead to the identification of so-called “hidden energy poor”.

One problem with this approach is that it is based on actual expenditures and is therefore subject to the preferences and behaviour of individual households, giving a biased result.

For this reason it is finally proposed an indicator based on estimates of heating demand according to geographical location and household composition (unaffected by household preferences).

A family is then considered to be in energy poverty ($FP_i=1$) if its total expenditure (S_i^{tot}), net of heating costs ($S_i^{heatmin}$, estimated by the model, and not based on the family's actual expenditure), is below the relative poverty threshold provided by ISTAT (σ).

$$FP_i = I[(S_i^{tot} - S_i^{heatmin}) < \sigma] \quad (16)$$

Fina [18] focus instead on the redistribution of the costs for running the REC, looking at the income level of the users. They first distinguish between three types of income level: low-income households, medium-income household and high-income households.

The redistribution mechanism is later set up such that the total EC-incurred grid costs of low-income participants are being borne by medium- and high-income households.

Based on Comparison with non-cooperative situation

Some RECs adopt a redistribution logic based on the overall community savings, i.e. the gain that the RECs collectively obtain compared to the non-cooperation situation.

Stentati et al. [24] propose a fair distribution in order to ensure that all members don't increase their bills in comparison with individual status and that no one will find advantageous exiting the community.

To ensure this, incentives are redistributed in this way:

- If an entity has a profit in the community worse than acting individually, it receives an initial compensation which fills the gap.
- The remaining portion of the incentive is divided among the entities proportionally to their contribution to the community self-consumption.

The total incentives (I^{CC}) are hence divided into a first tranche which is intended to compensate those who “lose out” from joining the REC (I_{comp}^{CC}), and a second tranche for redistribution according to contribution (I_{red}^{CC}), as in the formula below.

$$I^{CC} = I_{comp}^{CC} + I_{red}^{CC} \quad (17)$$

In the paper by Volpato et al. [25], instead, it is introduced the presence of an aggregator within the REC to manage members' bills and redistribute them.

In this case energy bills of the communities are considered as part of the revenues the energy community need to distribute. The idea behind the mechanism is trying to encourage those members who have the higher advantage in becoming independent prosumers from the status of consumers, to become part of the EC. The final bill of each prosumer belonging to the EC $b_{p,new}$ is calculated according to the following formula:

$$b_{p,new} = b_p - (income_{EC} - costs_{EC}) * \frac{(b_{p,0} - b_{p,1})}{\sum_{p=1}^n (b_{p,0} - b_{p,1})} \quad (18)$$

where:

- b_p is the bill of the prosumer p within the EC before allocation;
- $income_{EC}$ is the sum of all the electricity bills (b_p) before allocation;
- $costs_{EC}$ are the total costs (i.e., the objective function to minimize);
- $(b_{p,0} - b_{p,1})$ is the economic saving of the prosumer p achieved by moving from the status of pure consumer (having the total electrical demand fulfilled by the grid and the bill $b_{p,0}$) to the status of independent prosumer, i.e., not belonging to the EC (having the bill $b_{p,1}$ lower than $b_{p,0}$);
- n is the number of prosumers.

Equation 17 subtracts from the bill b_p (before the distribution of revenues) a part of the net revenues of the EC, proportionally to the economic savings achieved in changing the status from pure consumer to that of independent prosumer.

2.2.4. Benefits allocation through dynamic pricing

In contrast to centralised ex-post redistribution mechanisms based on static or historical criteria, more dynamic and decentralised approaches are emerging, made possible by the development of digital technologies and smart grid infrastructure. In this context, models based on price dynamics, such as peer-to-peer (P2P) systems and Local Electricity Markets (LEM), are becoming increasingly important in several European countries (the Netherlands, the United Kingdom, France, Switzerland, Portugal), while in Italy they are still in their infancy.

These models go beyond the logic of static redistribution of benefits, introducing direct or semi-direct exchanges between energy actors (prosumers and consumers) within the community, with prices determined in real time based on production, demand and flexibility. This allows for a more efficient and performance-based allocation of economic value, as well as promoting greater user involvement in local energy management.

Recent literature offers different perspectives on these approaches.

For example, [26] propose a community P2P trading model that aims to balance the interests of consumers and producers through shared market rules, showing how such mechanisms can reduce energy costs and ensure a fair distribution of revenues.

Other contributions, such as those by [27] and [28], analyse different local price formation strategies in LEMs, highlighting how the choice of pricing mechanism influences the balance between producer and consumer benefits.

Finally, studies such as [29] and [30] explore ex-post allocation approaches and community management models (e.g. Energy Community Manager) aimed at collective cost optimisation and equitable energy distribution within the community.

Overall, these experiences suggest a transition towards more participatory and data-driven energy governance models, in which price flexibility and local cooperation are key elements for the economic and social sustainability of energy communities.

Although these are valid approaches to the allocation of economic benefits in RECs, we have decided to mention them but not to explore them in depth, given that our research has focused on methods of benefit redistribution, which therefore use an “ex post” approach.

2.3. Summary table

As an effective tool for summarising the analysis provided above, a table has been created summarising the main methods identified in the literature.

Six macro-categories have been identified, which are as follows:

- Game theory models
- Shapley approximation based models
- Simplified models
- Multicriteria approaches
- Multicriteria approaches with focus on Social Inclusion
- Comparative approaches

Each of the above methods belongs to one of these categories, and the pros and cons of each have been analysed.

Important information is the country in which that method was identified, as each country has adopted a rather diverse regulatory plan.

An essential piece of additional information provided by the table is the categorisation of the technical, economic and social parameters taken into consideration in the redistribution strategy of the various methods analysed.

This analysis was particularly important because it made it possible to identify the different parameters used in the literature and also allowed us to identify the gap that will be explained later.

The entire table is available in *Appendix A*.

2.4. Fairness estimation

In parallel with the analysis of the methods of redistribution of economic benefits, the literature discusses the concept of fairness, used to evaluate the allocation of the same methods mentioned above.

Different points of view and interpretations of what is fair emerge from the analysis of the literature. In the papers proposed by Casalicchio et al. [10], a fair distribution of the payoff among users depends on the allocation of a share which is proportional to their marginal contribution to the system.

A key point of the paper is that fair does not necessarily imply an equal distribution of the benefits. Sometimes, it is fair that a user is granted more benefits than another user; therefore, it is necessary to select the most appropriate allocation metric.

It is identified how the user contribution to the community can be quantified, specifically considering economic terms.

Following, the Fairness Index is proposed, which reflects how far is the proposed redistribution to the optimal configuration, found by allocating to each user benefits according to his Marginal Contribution.

$$FI(m) = \begin{cases} \frac{\sum_{i=1}^m |D_i - D_{cd,i}|}{\sum_{i=1}^m |D_{w,i} - D_{cd,i}|} & \text{if } m = m_{tot} \\ m_{tot} - m & \text{if } m \neq m_{tot} \end{cases} \quad (19)$$

Where:

D_i = distribution percentage of benefits for member i

$D_{cd,i}$ = contribution distribution percentage of benefits for member i

$D_{w,i}$ = worse distribution percentage of benefits, according to the contribution distribution, for member i

m = number of members for which $D > 0$

m_{tot} = total number of members in the EC

The fairness index (FI) is applicable to any EC regardless of the size and it returns a value between 0 and 1 if each individual member has a profit: the closer the value is to zero the fairer and the more suitable is the applied redistribution method. If some participants do not get any benefit—that is, they have more expenses than benefits, according to the adopted distribution—the distribution will have negative terms and the index will return the number of unhappy users. The integer represents the number of unhappy members.

The denominator of the FI involves the introduction of an additional benefit distribution called D_w which is the furthest from the optimal one.

In the paper by Gasca et al. [31] a more detailed analysis is provided. With reference to the P2P context, for example, and citing the paper by Perger [32] reference is made to the case in which the willingness to pay for a preferred user is taken into account in the distribution of internal energy.

The research further compares four different fairness indexes and apply them to the context of ECs.

The first considered method relies on the Jain Index, designed to achieve equal distribution of resources, hence, to ensure equity.

The MinMax and QoE indexes measure fairness by comparing the highest and lowest bill savings in a community, and by comparing the standard deviation of them, respectively.

The last analyzed was the Distance Index, which is similar to the previously cited Fairness Index, because it compares a redistribution method with one based on Shapley value mechanism.

Volpato et al. [8] discuss about fairness in relation to each specific Mathematical model used while redistributing economic benefits. More specifically, “individual fairness” is the one guaranteed by the Shapley Value based method, since it allows for the allocation of higher profits or lower costs to members of the EC who contribute more to increasing the economic benefits of the coalitions of the EC. The Nucleolus approach instead guarantees “collective fairness”, since it assigns to each coalition of users more economic benefits inside the EC, than the case without it. Last, the notion of “proportional fairness” is described in relation to the adoption of Nash Bargaining optimization approach, allowing for the simultaneous optimization of the total benefit of the EC and the individual benefits of its members.

2.5. Gap identification and Research Questions

A comprehensive review of the literature on the utilisation and redistribution of the economic benefits from RECs revealed areas that have yet to be furtherly investigated.

Specific recommendations for future research emerged in various papers , such as the one by Basilico et al., [33], concerning the need to integrate “technical, economic and social considerations for a comprehensive and holistic understanding” of how redistribution of economic benefits could be done. Many papers develop methods that analyse one of these dimensions individually, but they fail in considering them as a whole in order to identify an appropriate allocation strategy. As Trevisan states [20] additional parameters are needed to consider a broader community context or redistribution metrics based on other KPIs that consider several beneficial contributions to the energy community. Further research into the social dimensions of REC , also, could yield valuable insights into strategies for encouraging their widespread adoption. Especially regarding the utilization of appropriate measures to

identify vulnerable users and the way to include them in a more fair redistribution strategy.

Diverse factors can be considered during a fairness estimation, including investment, assets ownership, user types and characteristics, operational decisions, and cost allocation methods. For this reason, refining the parameters of the benefits distribution models, particularly by enhancing the mechanisms for rewarding virtuous behaviour and addressing income inequality, has been repeatedly recognised as a topic of great importance in making the REC more fruitful for its users and more attractive to the new users [31].

This is essential to ensure that each type of member within the REC should be allocated a portion of the benefits that does not generate discontent, guaranteeing as much economical stability as possible, discouraging members from leaving and more importantly, make joining REC as appealing as possible to new users, so as to increase REC diffusion and spread.

The thesis will seek to contribute to scientific literature by exploring how it is possible to develop a strategy for redistributing benefits which, while complying with regulatory constraints, manages to guarantee a model of fairness and effectiveness capable of promoting the spread of RECs in the country.

From these considerations, the research questions that this thesis will attempt to answer are as follows:

Which technical, economical and social parameters need to be included in the benefits redistribution strategy to meet the needs of each category of members and ensure that each member is allocated the incentive share that reflects their contribution to the REC?

4 Experimental model definition

After analysing academic literature, the Italian regulatory framework and numerous case studies from RECs currently operating in Italy, the next step in my thesis was to develop my own model proposal that would take into account the gaps identified in the literature and, at the same time, integrate perfectly with the constraints and directives imposed by the Italian regulatory framework.

This was not only a fundamental part of my thesis, but also an important step in my curricular internship at ENEA, where I was able to contribute to the in-depth study of the redistribution of economic benefits.

The model was constructed using MatLab software, supported by RECON [39] provided free of charge by ENEA.

RECON is a web application developed by ENEA and aimed at supporting preliminary energy, economic, and financial assessments for the establishment of renewable energy communities or groups of renewable energy self-consumers acting collectively.

It is updated to the current legislative and regulatory framework in Italy and allows for the analysis and simulation of a REC's behavior over time through parameters such as user withdrawal and injection profiles and financing methods for the plants. RECON calculates, among other things, the yield of production plants, physical self-consumption (for each prosumer), and distributed self-consumption within the configuration.

The development of the model followed four main phases, which will be briefly mentioned in this introductory chapter and analysed in greater depth later on.

The first phase of the model represented an “a priori” step with respect to the actual distribution of economic benefits. Its main objective was to simulate the calculation of the incentive made by the GSE, with the ultimate aim of obtaining the incentive associated with each plant in the configuration.

It also provided an important distinction between the incentive coming from plants that benefited from a capital grant (whether NRRP or other European grants) and plants that did not, and between the excess incentive quota (and therefore bound to comply with certain destinations, as indicated by the GSE) and the non-excess quota.

In the second phase, a redistribution methodology proposal was provided, based on the outputs from the first phase.

In this phase, important steps had to be taken into account, such as REC expense coverage, the possibility of provisions, and, lastly, the management of allocation methods within user categories.

The output of this phase was the allocation of the incentive quota to individual producers, consumers, and prosumers, as well as to social purposes.

The third phase was a validation phase of the proposed model. For this validation, the presence of prosumers within the REC was introduced as a simulation. This was followed by an analysis of the incentive values for the 20 years of coverage guaranteed by the state.

In this phase, the results of the model were compared with data from the RECON economic simulator to ensure the accuracy of the model.

In the fourth and final phase, a series of REC case studies, created specifically using the RECON economic simulator, were analysed.

The analysis of these scenarios was useful in understanding how redistribution varied even in cases that differed significantly in various aspects, such as the composition of REC users, widespread self-consumption values, and the presence or absence of businesses within the community.

4.1. Calculation of fundamental quantities

The model established starting points to be used as input data. These starting points refer to a series of data provided by the RECON simulator, and are as follows:

The first fundamental indication as a starting point for calculating the incentive was a table with the set of PODs characterising an energy community.

For each POD, the user category (SME, private citizen, third sector entity, association and all other categories of users who can take part in a REC) is identified, as well as the user's role in the configuration (producer, prosumer or consumer) and other more specific information, such as specific indications on the trend in consumption associated with consumers and prosumers.

Another table that is very useful for redistribution purposes is the table referring to the REC plants analysed. For each plant, important information has been provided, such as: plant power and date of commissioning, as well as information specifically relating to the plant's economic data, such as the plant's access to NRRP or European subsidies.

Fundamental information also includes data relating to production, consumption, withdrawal and injection by each user in the REC. This data referred to each time slot of the period in which the incentive is actually guaranteed.

The table showing the value of the zonal price over time was also relevant for calculating the hourly TIP.

The first step taken to calculate the final incentive was to identify the shared energy for each hour associated with each plant.

Shared energy was calculated as defined, i.e. as the minimum between energy fed into the grid and energy withdrawn by users.

The association with each plant followed the GSE calculation logic, which gives priority in the allocation of shared energy to plants that entered into operation first.

The second step involves calculating the share of shared energy associated with each plant to which the reduction factor F applies or does not apply. As specified in the GSE's operating rules, the factor F : “does not apply to eligible shared electricity (E_{ACI}) relating to withdrawal points owned by local authorities, religious bodies, third sector and environmental protection bodies, and natural persons”.

This breakdown was made by categorising the individual users and dividing the total withdrawal into withdrawals relating to exempt and non-exempt categories.

Subsequently, the TIP associated with each plant was calculated.

The calculation formula is the same as that provided by the GSE operating rules, and is shown below.

$$TIP_h = \{min[CAP; TP_{base} + max(0; 180 - Pz)] + FC_{zonal}\} * (1 - F) \quad (32)$$

As mentioned above, the information for calculating the zonal price or other parameters (such as FC_{zonal} or TP_{base}) was provided by the RECON simulator.

Once the share of exempt and non-exempt shared energy had been established and the TIP calculated for each plant, it was possible to identify the incentive associated with each plant using the formula below, and then subsequently find the total value of the incentive accrued by the REC in that year.

$$Inc_i = SE_{nonF} * TIP_i + SE_F * TIP_i * (1 - F) \quad (33)$$

In the formula above, SE_{nonF} refers to the amount of shared energy associated to that plant to which does not apply the reduction linked to the F parameter; SE_F refers instead to the remaining part of the shared energy associated with that plant. TIP_i is the feed-in premium tariff associated to that plant.

The final part of the first phase of the model aimed to identify the share of incentives exceeding the shared energy threshold value and those not exceeding it. This refers to the fact that, as specified in the GSE operating rules, if shared energy exceeds 55% (45% if we are referring to plants benefiting from a capital grant) of the total energy fed into the grid by the plants on an annual basis, the incentive quota relating to that excess

portion must necessarily be allocated to non-business consumers or used for social purposes that have an impact on the areas where the sharing plants are located.

This calculation was made based on the data obtained previously, to arrive at the end of the first phase of the model with the following output: the bound and unbound incentive quota for plants that have benefited from non-repayable grants and the bound and unbound incentive quota for plants that have not benefited from any non-repayable grants.

To verify the accuracy of the incentive calculation algorithm and the incentive value assigned to individual plants, the final values obtained from applying the model were compared with those from the RECON software simulation.

This phase was useful for correcting and optimising certain steps in the “initial” version of the model and ensuring that we had solid data on which to base phase 2, i.e. the redistribution strategy for individual users.

4.2. Implementation of the allocation Strategy

Once the first phase of preparing the algorithm for calculating the total REC incentive and the incentive associated with individual plants was completed, the next step was to work out a redistribution of the incentive within the REC.

It is important to note that the initial data was based on a rather simple REC model, consisting of four producers and eight consumers, whose relevant data (especially withdrawals and feed-ins) was obtained from the RECON simulator.

As mentioned above, the input data for this phase were the total incentive values associated with each plant. Another important piece of information obtained from this phase was the proportion of the incentive that was bound and unbound.

A first decision point imposed by redistribution was the coverage of expenses. As indicated by Italian regulations, the unbound incentive quota was taken as the starting point for covering the expenses incurred by the REC. This data was arbitrarily included as the initial parameter of the model, effectively being an “external” parameter.

In order to enter a parameter that was as accurate as possible, the guidelines provided by ART-ER and the EMILIA ROMAGNA REGION [40] were taken as a reference point, within the 'Methodological guidelines for the preparation of feasibility studies for the establishment of CERs', the fourth notebook for supporting the energy transition of REC and renewable energy self-consumers.

The cost items analysed are those relating to operating costs, namely:

- Plant management and maintenance costs
- Legal entity management costs

- Configuration management costs
- GSE administrative costs

Once the expenses had been deducted from the uncommitted portion of the incentive, it was necessary to analyse how to structure the division of the incentive. The main question was whether to consider separately the two portions relating to the incentive obtained from plants benefiting from the subsidy and the incentive obtained from plants not benefiting from the subsidy, which therefore did not give rise to a reduction in the subsidy.

The final decision was to keep the two quotas separate and proceed with the division of the incentive between the individual categories distinctly.

The main reason that led me to consider the two quotas separately was the fact that a plant that receives a capital grant, whether from Italian state grants (typically NRRP) or other European funds, contributes to a final reduction in the incentive.

Following this logic, it was decided that it was appropriate to leave it up to the individual RECs to decide how to treat the two quotas separately, and whether to distinguish the allocation to plants that have received state or European subsidies, contributing to a reduction in the total redistributable incentive, from plants that are instead associated with the total (entire) incentive quota.

Having decided to separate the two quotas net of costs, it was necessary to further allocate the net unconstrained incentive. In this regard, two allocation methods were identified: the first uses the entire incentive quota as the basis for allocating the two net uncommitted incentive quotas. The second, on the other hand, considers the final values of the uncommitted incentive associated with each of the two categories (exempt and non-exempt plants) as the basis for calculating the net incentive associated with one category rather than the other.

Within the model, it was decided to use various simulations to analyse how much these two different scenarios could actually change the redistribution to individual users. Therefore, an 'if' statement was used in each simulation to analyse the distributions to individuals using both methodologies.

The results of these analyses will be discussed in the section related to the simulations and scenario analysis.

Given the reduced and unreduced net unbound incentive values, and given the total bound incentive value, the next step was to allocate a reference percentage of the quota to each user category.

As regards the bound quota, in accordance with regulatory requirements, the only division to be identified was between the percentage to be allocated to non-business consumers and the percentage to be allocated to activities or services with social purposes.

As regards the unbound quota, however, the allocation was more complex. In this case, it was necessary not only to identify the percentage to be allocated to each category, but also to distinguish between the two cases of reduced and non-reduced tariffs.

The uncommitted incentive quota can be used for various purposes: the first is as a reserve/provision for the CER, both to cover any extra expenses and to have cash available to guarantee the security of the REC, as well as to have a sum available for future investments by the REC itself.

The second is to guarantee compensation to those who have actively contributed to the energy fed into the grid. Similarly, it guarantees compensation to those who have actively contributed to the withdrawal of energy from the grid.

A final consideration is social impact: REC may decide to allocate not only part of the regulated tariff but also part of the unregulated tariff to activities or services with a social impact.

In the initial phase of the model, it was decided to leave the definition of these allocation percentages to the individual RECs, ensuring that each one could define its own strategy by modifying the various percentages to be allocated to the individual categories.

While leaving these parameters as data to be entered externally, for the analysis of the various simulations it was decided to rely on information obtained from the analysis of REC regulations to identify a “symbolic value”, in particular for the allocation of the incentive not constrained to be given to producers and consumers.

An important consideration regarding the model is the management of the ‘prosumers’ category within the breakdown.

As the model is based on hourly data, prosumers are identified every hour either as ‘consumers’, if the final balance between withdrawal and injection is positive, or as ‘producers’, if the final hourly balance between withdrawal and injection is negative. For this reason, when calculating the total incentive to be allocated to individual prosumers, part of it will come from the share received thanks to shared energy withdrawal (based on the hourly bands in which the prosumer is categorised as a consumer), while the other part will come from the share due to them from their contribution to the injection of shared energy into the REC.

4.2.1. Allocation to single members

A fundamental part of the redistribution model is the management of incentive allocation to individual producers and consumers within the two categories.

As previously mentioned, after a series of considerations regarding the distinction between reduced and non-reduced net incentives, it was decided to keep the two quotas separate. Consequently, the allocation to individual producers and consumers

will be based on the unrestricted incentive allocated to each of the two categories (reduced and non-reduced), and redistribution will take place separately between the two. It goes without saying that producers who have not received any capital grants, and therefore have not contributed to the reduction in the quota, will receive the full amount of the incentive. Producers who have received capital grants, on the other hand, will receive a reduced amount of the incentive.

Starting with the first of the two categories, i.e. producers, an initial option considered was to use the calculation algorithm provided by the GSE: following its logic, each individual producer would be allocated a share of the incentive that they had helped to generate.

However, this logic has a fundamental problem: it tends to significantly favour producers with older plants, which would not only be counterintuitive, effectively rewarding producers with plants that are less efficient and less innovative than newer ones, but would also have the major problem of discouraging new potential producers/prosumers from entering the REC.

This would be particularly serious in the event that a REC is unbalanced on the withdrawal side and requires further stimulus for the injection of energy into the grid for sharing purposes.

Furthermore, from the point of view of fairness, this would perhaps disadvantage producers with newer plants that have greater power and which, despite contributing significantly to the injection into the grid, would be allocated only a small part of the incentive.

Once this logic was discarded, the focus shifted to defining parameters that could best reflect the producer's contribution to the final incentive obtained by the REC.

Linked to this, it was necessary to identify the most appropriate time frame for our purposes. With hourly data on the inputs of individual producers available, it was deemed appropriate to start from individual time slots in order to obtain information useful for identifying the most accurate allocation criterion possible.

In this regard, two main parameters were identified:

- Hourly energy fed in by each producer: both in academic literature and in the empirical data analysed, the energy fed in by each producer is considered a good proxy for the contribution made by the individual producer.
- Hourly shared energy: in the proposal to reallocate the incentive to individual producers, it was thought that simple energy input was not sufficient, but that a parameter needed to be introduced that would better reflect how much a producer had actually contributed.

The basic idea is to reward producers who have made a large contribution to shared energy, especially at times when the latter was high and contributed most to the incentive.

To better explain the concept, here is an example: a producer who has fed a large amount of energy into the grid, but only during hours when consumption by other grid users was minimal, has not made a significant contribution to the REC in terms of increasing the incentive. A producer who, on the other hand, fed a certain amount of energy (perhaps even the same amount) into the grid at times when consumption by other users was high contributed much more to the final incentive generated by the REC.

The final formula representing the contribution of each individual consumer first considers the contribution to the grid by each individual producer on an hourly basis. It then considers the value of shared energy at that given time, in order to give an effective weighting to the contribution in question.

Once this has been done for every hour of the year, the annual incentive allocated to each individual producer is given by the previously calculated annual contribution.

Below is a formula for each of the two reference rates (reduced rate and non-reduced rate).

Producers with contributions:

$$Inc_{i,contr} = Inc_{NoRestr,contr} * \%Prod_{contr} * \frac{\sum_y \frac{E_{inj_{i,y}}}{\sum_i E_{inj_{i,y}}} * SE_{contr,y}}{\sum_i (\sum_y \frac{E_{inj_{i,y}}}{\sum_i E_{inj_{i,y}}} * SE_{contr,y})} \quad (34)$$

Being:

i : the i -th producer receiving grants from State or from external sources

y : the y -th hour (from 1 to 8784, hours from a leap year)

$Inc_{i,contr}$: the final incentive allocated to the i producer/prosumer for his contribution

$Inc_{NoRestr,contr}$: the net non restricted incentive, associated to plants whose owners have received grants from the State or from external sources

$\%Prod_{contr}$: the allocation percentage to producers/prosumers who have received grants from the State or from external sources

$SE_{contr,y}$: the total shared energy associated with plants whose owners have received grants from the State or from external sources, at hour i

$E_{inj,i,y}$: the energy injection by the producers i , at hour y

Producers with no contributions:

$$Inc_{i,Nocontr} = Inc_{NoRestr,Nocontr} * \%Prod_{Nocontr} * \frac{\sum_y \left(\frac{E_{inj,i,y}}{\sum_i E_{inj,i,y}} * SE_{Nocontr,y} \right)}{\sum_i \left(\sum_y \frac{E_{inj,i,y}}{\sum_i E_{inj,i,y}} * SE_{Nocontr,y} \right)} \quad (35)$$

Being:

i : the i -th producer not receiving grants from the State or from external sources

y : the y -th hour (from 1 to 8784, hours from a leap year)

$Inc_{i,Nocontr}$: the final incentive allocated to the i producer/prosumer for his contribution

$Inc_{NoRestr,Nocontr}$: the net non restricted incentive, associated to plants whose owners have not received grants from the State or from external sources

$\%Prod_{Nocontr}$: the allocation percentage to producers/prosumers who have not received grants from the State or from external sources

$SE_{Nocontr,y}$: the total shared energy associated with plants whose owners have received grants from the State or from external sources, at hour i

$E_{inj,i,y}$: the energy injection by the producers i , at hour y

The next step was to identify strategies that were as fair as possible for allocating the incentive to individual consumers.

As already done for producers, the basic idea was to consider those parameters that were necessary to identify each consumer's contribution to the incentive:

- Hourly energy withdrawn: as with the energy fed in by producers, for consumers, the energy withdrawn is a good proxy for the consumer's contribution within the REC.
- Hourly shared energy: here too, it is necessary to distinguish between the hours when shared energy was present, and therefore part of the incentive was generated.

Using these two parameters, two approaches for allocation to individual consumers were identified.

The first, called the “Proportional Approach”, consists of considering the hourly energy withdrawn by the individual consumer only at times when the shared energy is greater than 0.

The aim is both to identify the consumer's consumption and to limit the scope to those time slots when there was actually energy fed into the grid by REC plants.

The main purpose of this mechanism is to reward those consumers who have consumed energy when it has actually been fed into the grid from renewable sources. At that point, much more value is given to a certain amount of energy consumed during peak hours of the day than to the same amount consumed during the evening or night hours, when feed-in is minimal or zero.

The second approach to allocating the incentive to individual consumers is based on the hourly allocation of shared energy to individual consumers. In fact, if in cases where shared energy is determined by the energy withdrawn by consumers (lower than that fed in), the allocation of shared energy to individual consumers is fairly clear, as each consumer is assigned their own withdrawal, in cases where shared energy is determined by the energy fed in (lower than the energy withdrawn), it is not so easy to determine how to allocate that shared energy to individual consumers.

The name of this strategy is “Layered Approach”, and it is based on a shared energy allocation methodology already proposed in the literature by Minuto et al. and analysed in the chapter on “Literature review”.

The ‘Layered’ algorithm provides a fixed guaranteed allocation quota for each member, whereby a small portion of the shared energy quota (which tends to coincide with the minimum withdrawal among users) is distributed to each user. The same applies to the remaining energy quota: looking at the ‘residual’ shared energy still to be allocated to each consumer, it allocates to each individual the minimum of the remaining residual energy, and if this is not possible, it allocates the total remaining shared energy, divided equally among all consumers who have not been guaranteed the entire shared energy quota.

The name ‘Layered’ derives precisely from the fact that it is an algorithm that works in ‘layers’. Unlike the proportional approach, this approach provides greater protection for consumers who have a low level of withdrawals, such as individuals.

Once the share of shared energy has been allocated to each consumer on an hourly basis, the annual allocation is made by looking at the amount of annual shared energy associated with each consumer.

It is important to emphasise that these two algorithms are to be applied separately for the two quotas, restricted and unrestricted.

This distinction is necessary because the restricted quota can only be allocated (in addition to social purposes) to non-business consumers, while the unrestricted quota can be allocated to all categories of consumers.

For the final calculation of the incentive to be allocated to individual consumers, therefore, a distinction will be made between two cases: business consumers and non-business consumers.

Business consumers:

$$Inc_i = (Inc_{NoRestr,Nocontr} * \%Cons_{Nocontr} + Inc_{NoRestr,contr} * \%Cons_{contr}) * share_i \quad (36)$$

Being:

i : the i -th business consumers

Inc_i : the incentive allocated to i -th consumer

$Inc_{NoRestr,Nocontr}$: the net non restricted incentive, associated to plants whose owners have not received grants from the State or from external sources

$\%Cons_{Nocontr}$: the allocation percentage to consumers(/prosumers) of the incentive coming from plants whose owners have not received grants from the State or from external sources

$Inc_{NoRestr,contr}$: the net non restricted incentive, associated to plants whose owners have received grants from the State or from external sources

$\%Cons_{contr}$: the allocation percentage to consumers(/prosumers) of the incentive coming from plants whose owners have received grants from the State or from external sources

$share_i$: Share of the total net incentive allocated to consumers to be allocated to individual consumers using the Proportional approach or the Layered approach

Non-business consumers:

$$Inc_i = (Inc_{NoRestr,Nocontr} * \%Cons_{Nocontr} + Inc_{NoRestr,contr} * \%Cons_{contr}) * share_{i,NoRestr} + Inc_{Restr} * \%Cons_{Restr} * share_{i,Restr} \quad (37)$$

Being:

i : the i -th non-business consumers

Inc_i : the incentive allocated to i -th consumer

$Inc_{NoRestr,Nocontr}$: the net non restricted incentive, associated to plants whose owners have not received grants from the State or from external sources

$\%Cons_{Nocontr}$: the allocation percentage to consumers(/prosumers) of the incentive coming from plants whose owners have not received grants from the State or from external sources

$Inc_{NoRestr,contr}$: the net non restricted incentive, associated to plants whose owners have received grants from the State or from external sources

$\%Cons_{contr}$: the allocation percentage to consumers(/prosumers) of the incentive coming from plants whose owners have received grants from the State or from external sources

Inc_{Restr} : Total Restricted Incentive

$\%Cons_{Restr}$: The allocation percentage to consumers (non-business consumers) of the total restricted incentive

$share_{i,NoRestr}$: Share of the net non-restricted incentive allocated to consumers to be allocated to individual consumers using the Proportional approach or the Layered approach

$share_{i,Restr}$: Share of the total restricted incentive allocated to consumers to be allocated to individual consumers using the Proportional approach or the Layered approach

As can be seen from the formulas, business consumers are allocated only part of the unrestricted incentive, while non-business consumers, in addition to the part of the unrestricted incentive to which they are entitled, are also allocated part of the committed incentive, as they are plausible recipients of the restricted portion.

Given the complexity involved in dividing the incentive and the various quotas associated with each category, a block diagram summarising the various steps of the proposed model is given below.

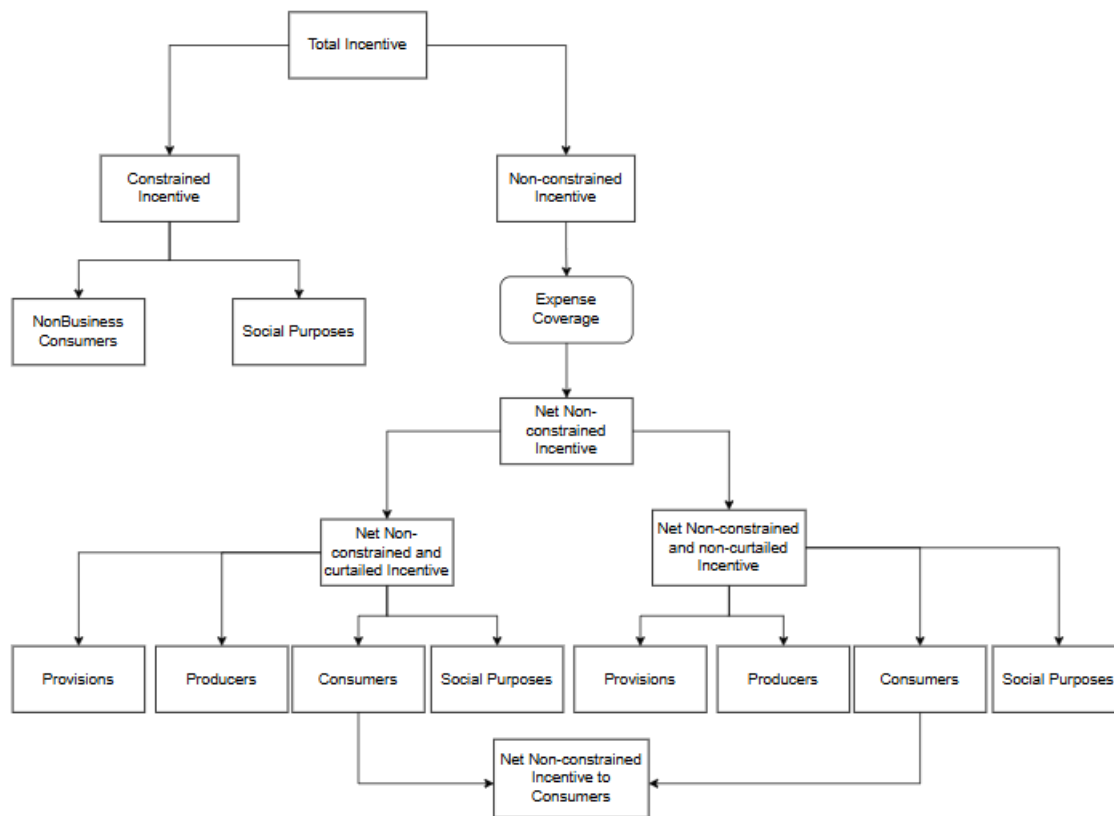


Figure 21: Structure of the proposed model

4.3. Algorithm testing

The third phase of model development involved testing the MatLab algorithm by checking the results.

This was done by analysing the incentive value assigned to each plant and then the total incentive value accrued by the REC, comparing the final values resulting from the simulation using RECON with the results of the newly constructed model.

After an initial phase in which these did not match, the problem was solved by adjusting the energy allocation algorithm relating to the withdrawal of users exempt from reduction. Priority was given first to plants benefitting from subsidies, ordered chronologically, and then to plants not benefitting from subsidies, also in chronological order.

The next step was to extend the MatLab algorithm to cover the entire period during which the REC incentive was guaranteed, i.e. the first 20 years. This was achieved with a for loop, used to cover from year 1, when the REC was created, to year 20, when the incentive support ends.

The comparison with the data provided by RECON was also crucial at this stage, ensuring consistency between the values given by the algorithm and those obtained by RECON for the entire duration of the period analysed.

A final confirmation and validation of the algorithm used was to feed the proposed algorithm with a case study that included the prosumer category within the model.

In ‘Prova1’, the first REC simulation provided to the model, this specific category had not been taken into consideration.

After its inclusion in the ‘Prova2’ simulation, it was possible to identify the incentive associated with prosumers as well.

As mentioned above, the total incentive allocated to individual prosumers is given by the sum of the incentive derived from the contribution to shared energy input and the contribution to energy withdrawal.

4.4. Simulation and scenario analysis

Once the model had been created and validated, it was decided to proceed with a series of simulations and scenario analyses. There were several reasons for this: first of all, to test the model and its behaviour in various REC configurations, but also to analyse and distinguish between the various redistribution strategies analysed, such as the post-coverage distinction between unrestricted net incentives for plants without capital grants and unrestricted net incentives for plants with capital grants, or the redistribution of the incentive to individual consumers, following the “Proportional” approach or the “Layered” approach.

A total of four scenarios were analysed, each with a different configuration in terms of the number of consumers, producers and prosumers. In particular, in order to analyse “special” cases, the third simulation modelled an energy community with a high level of widespread self-consumption, while the fourth simulation modelled an energy community with a low level of widespread self-consumption.

The first two cases attempt to represent a typical REC, with widespread self-consumption levels of around 70%. The only difference between the two is that the second also includes prosumers, making the analysis more comprehensive.

In the following paragraphs, we will compare and analyse, for each scenario identified, the final allocation of the incentive to individual categories and individual users. In addition, we will analyse the different results obtained through the use of a redistribution that distinguishes the net incentive recognized for plants with subsidies from the net incentive released for plants without subsidies, and we will also compare the values allocated specifically to individual consumers, using the “Proportional” approach rather than the “Layered” approach, in order to understand the differences in behaviour between the two.

It is important to highlight the parameters used for the following simulations, which are kept constant for each one.

The zonal correction factor for calculating the TIP was set at 0.01 EUR/kWh, i.e. the factor corresponding to the Northern Italy region.

Following a review of the documentation provided by ART.ER and the EMILIA ROMAGNA REGION, total expenditure was estimated at 1,500 EUR/year.

Another important consideration must be made regarding the percentage value used during the simulations to simulate the choice of each REC with respect to the percentage allocation to each category of the exceeding quota, unrestricted quota with reference to plants benefiting from subsidies, and unbound quota with reference to plants not benefiting from subsidies.

Following a review of the empirical data from the analysis of the REC regulations, and our own partial modification, the allocation percentages used are as follows:

For the restricted quota:

- Allocation percentage to non-business consumers: 50%
- Allocation percentage for social purposes with an impact on the territories where the sharing facilities are located: 50%

For the net unrestricted quota allocated to plants without capital grants:

- Allocation percentage to producers: 50%
- Allocation percentage to consumers: 40%
- Allocation percentage for social purposes with an impact on the territories where the sharing facilities are located: 5%
- Allocation percentage to provisions: 5%

For the net unrestricted amount allocated to plants with capital grants:

- Allocation percentage to producers: 20%
- Allocation percentage to consumers: 60%
- Allocation percentage to social purposes: 10%
- Allocation percentage to provisions: 10%

One final piece of information needed to categorise the scenarios analysed is the fact that the breakdown between the net incentive allocated to plants with and without subsidies, after covering REC costs, was made on the basis of the gross unrestrained incentive share of each of the two categories.

With regard to the redistribution of incentives to consumers, the 'Proportional' approach was adopted in the following simulations.

4.4.1. Description of cases and redistribution to categories and users

The first simulated REC was composed as follows: eight consumers, including three small or medium-sized enterprises, and four producers. As mentioned above, a fairly

simple configuration was chosen for the first simulation, with the idea of modifying its components in subsequent simulations. The REC's widespread self-consumption values are around 67/68% (values coming from RECON simulation).

Below is a table summarizing the main inputs for the first simulation, including each user's category (the underscore '_'E' means Small or Medium Enterprises), annual withdrawals, installed power of the production unit (if any) and annual injections.

Table 2: Main inputs - first simulation

User#	Category	Annual Withdrawal [kWh]	Installed Power [kW]	Annual Injection [kWh]
1	Consumer	4040	0	0
2	Consumer	10958	0	0
3	Consumer	3127	0	0
4	Consumer	10958	0	0
5	Consumer	3127	0	0
6	Consumer	4040	0	0
7	Consumer_E	21916	0	0
8	Consumer_E	79723	0	0
9	Producer	0	8	9813
10	Producer	0	5	6133
11	Producer	0	30	36802
12	Producer_E	0	50	61337

One of the pieces of information provided by the model was the breakdown of the final incentive among the various categories: producers, consumers, social purposes and provisions.

The graph below shows the percentage of the total incentive allocated to each category. From this initial simulation, we obtain a redistribution that tends to favour consumers (who are allocated 40% of the total incentive) over producers (only 21% of the total incentive).

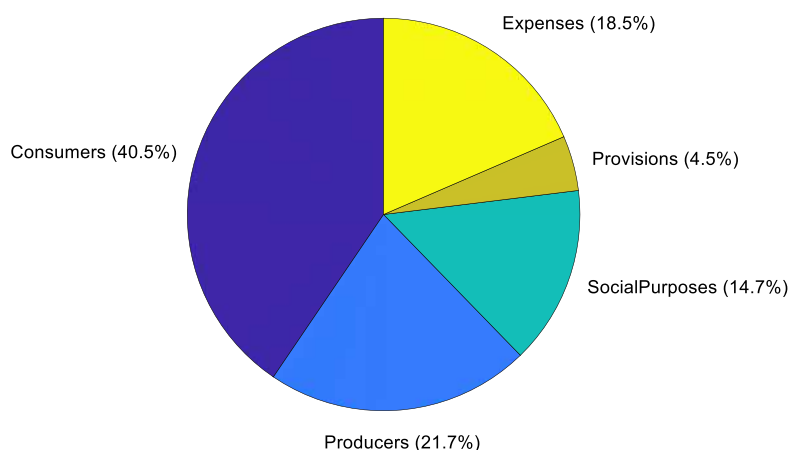


Figure 22: Incentive redistribution to categories – 1st simulation

The model not only guarantees the distinction between categories, but also allows the incentive quota allocated to each user to be identified.

As can be seen from the graph below, users are grouped into categories of producers and consumers (for prosumers, see the following simulations), and each user is assigned an “E” if they are a small or medium-sized enterprise. The total incentive quota allocated to each user is also divided into a “constrained” quota and an “unconstrained” quota.

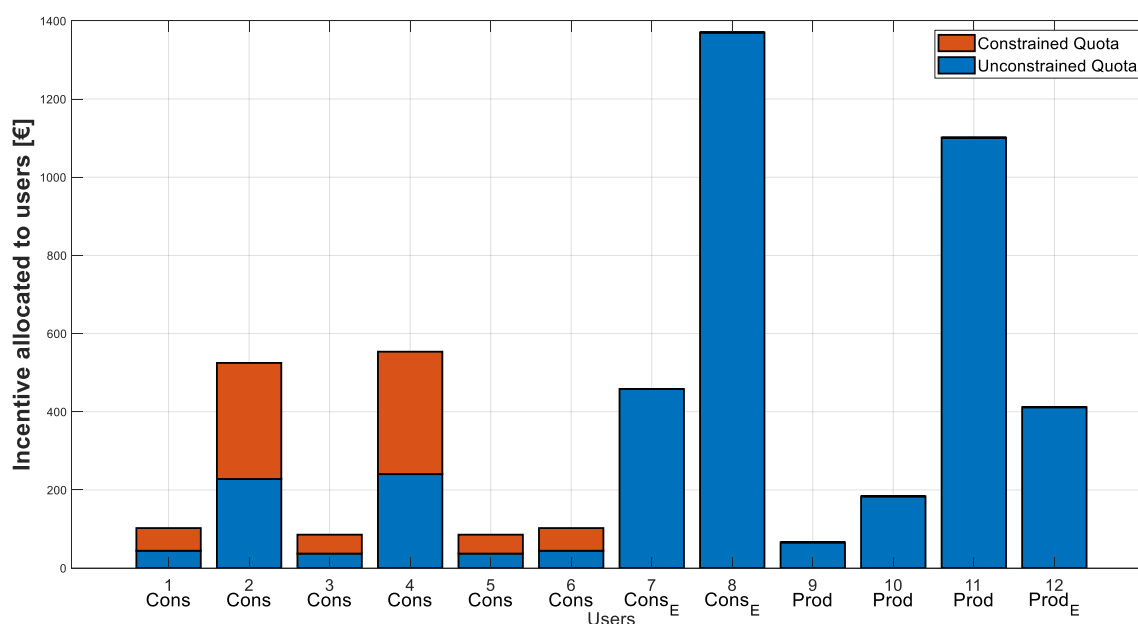


Figure 23: Incentive redistribution to users – 1st simulation

These distinctions are particularly relevant and significant because they show us how each quota varies for each user.

As can be seen from the graph, and defined by regulatory constraints, the ‘constrained’ quota is only present for non-business consumers.

The second simulation mirrored the first in terms of widespread self-consumption and consumer/producer balance, but introduced the category of prosumers into the model.

It consisted of eight consumers, including two SMEs, three prosumers and a single producer. The REC's widespread self-consumption values are around 63/64%.

Below is a table summarizing the main inputs for the second simulation.

Table 3: Main inputs - second simulation

User#	Category	Annual Withdrawal [kWh]	Installed Power [kW]	Annual Injection [kWh]
1	Consumer	4040	0	0
2	Consumer	3127	0	0
3	Consumer	10950	0	0
4	Consumer	10956	0	0
5	Consumer	21412	0	0
6	Consumer	4040	0	0
7	Consumer_E	31867	0	0
8	Consumer_E	119540	0	0
9	Prosumer	3003	40	41118
10	Prosumer	2405	10	10632
11	Prosumer_E	1792	70	75494
12	Producer	0	50	59836

The breakdown between categories now includes the category of prosumers, to whom approximately 27% of the total incentive is allocated. The presence of two “large” prosumers partially disadvantages individual producers, who receive only 3.6% of the total incentive. A large portion of the incentive is allocated to consumers, specifically

40%, at the expense of the social purposes category, which is slightly lower than in the first example. This may be mainly due to the fact that self-consumption in the second REC is a few percentage points lower than in the first, partially reducing the unallocated incentive quota and limiting the benefits allocated to social purposes.

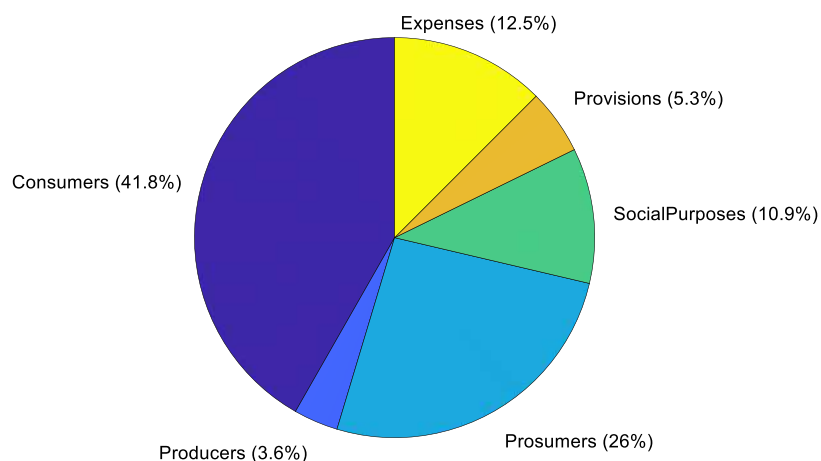


Figure 24: Incentive redistribution to categories - 2nd simulation

As regards the redistribution assigned to individual users, the trend does not vary particularly from the first case, with the exception of the only SME prosumer in the configuration, which receives most of the incentive, partly due to the size of the plant.

A feature similar to the first configuration is the presence of a 'strong' industrial consumer that attracts most of the unallocated quota, mainly thanks to its significant withdrawals during periods of high solar exposure.

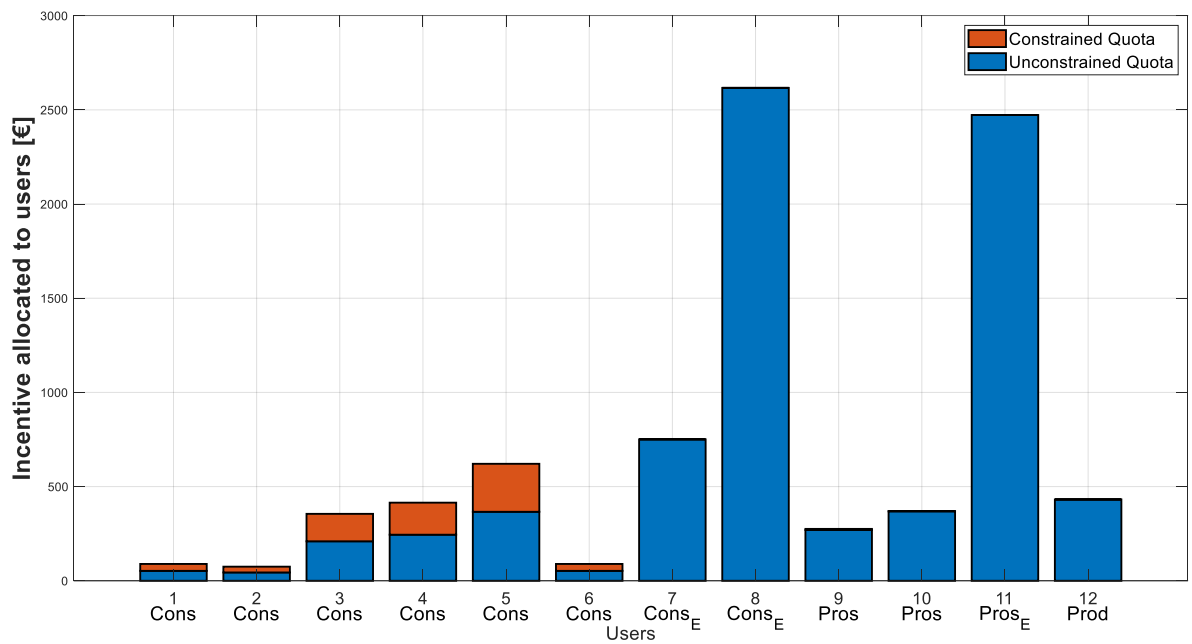


Figure 25: Incentive redistribution to users - 2nd simulation

The third simulated REC was deliberately constructed to have a high rate of widespread self-consumption (90%) in order to understand how the model would behave with these values and how the allocation dynamics would vary with such a high level of self-consumption. For this to happen, it was necessary to set up the configuration with a large number of consumers, whose consumption would cover the injection into the grid as much as possible.

The final configuration consisted of thirteen consumers (including five businesses), two prosumers and one producer.

Below is a table summarizing the main inputs for the third simulation.

Table 4: Main inputs - third simulation

User#	Category	Annual Withdrawal [kWh]	Installed Power [kW]	Annual Injection [kWh]
1	Consumer	3127	0	0
2	Consumer	99660	0	0
3	Consumer	110526	0	0
4	Consumer	79728	0	0
5	Consumer	4040	0	0
6	Consumer	69780	0	0
7	Consumer	4040	0	0
8	Consumer	74763	0	0
9	Consumer_E	84706	0	0
10	Consumer_E	139463	0	0
11	Consumer_E	119547	0	0
12	Consumer_E	59790	0	0
13	Consumer_E	84706	0	0
14	Prosumer	1776	10	10814
15	Prosumer_E	118251	70	24813
16	Producer_E	0	50	59836

An immediate observation when looking at the pie chart showing the breakdown of categories is an increase in the share allocated to social purposes (22%). This is easy to understand, considering the significant increase in the surplus share due to the high rate of widespread self-consumption. As the percentages for the individual categories in the redistribution strategy have not changed, the final share allocated to consumers does not vary significantly. Producers and prosumers see their percentage share shrink, again due to the significant increase in the surplus share.

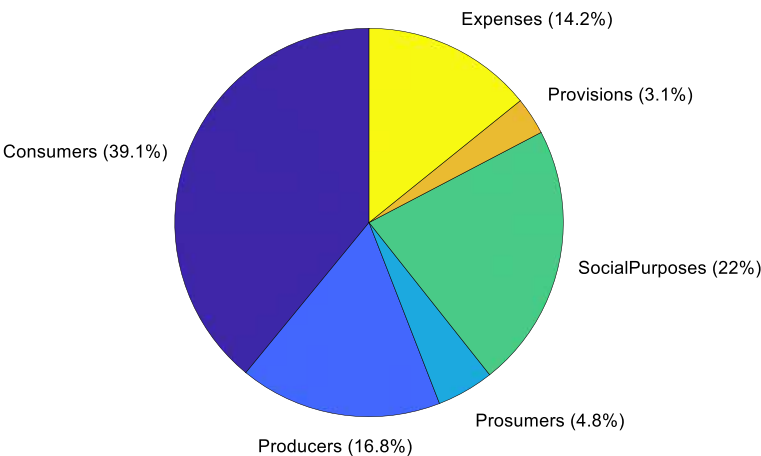


Figure 26: Incentive redistribution to categories - 3rd simulation

Looking at the allocation at individual user level, it can be seen that, compared to previous cases, non-business consumers significantly outnumber business consumers, precisely because of the significant share of the incentive allocated to them.

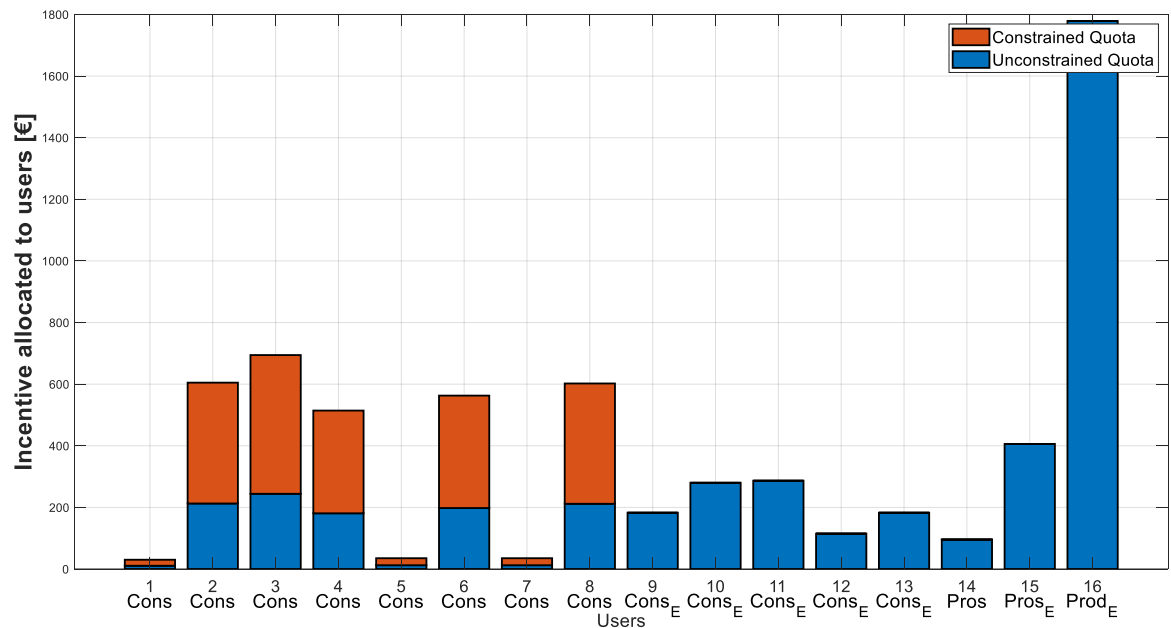


Figure 27: Incentive redistribution to users - 3rd simulation

The last simulation carried out was intended to consider the opposite situation to the previous one, i.e. a REC with a minimum level of self-consumption. For this reason, the REC was specifically constructed with an imbalance on the input side, consisting of six producers, three prosumers and only four consumers. The level of widespread self-consumption achieved in this case was 14-15%.

Below is a table summarizing the main inputs for the fourth simulation.

Table 5: Main inputs - fourth simulation

User#	Category	Annual Withdrawal [kWh]	Installed Power [kW]	Annual Injection [kWh]
1	Consumer	3127	0	0
2	Consumer	3127	0	0
3	Consumer_E	16934	0	0
4	Consumer_E	39866	0	0
5	Prosumer	1364	10	11195
6	Prosumer	3559	40	34845
7	Prosumer	3815	30	23158
8	Producer	0	10	12165
9	Producer	0	50	58141
10	Producer	0	10	11799
11	Producer	0	40	37031
12	Producer_E	0	60	73604
13	Producer_E	0	80	89525

Analysing the destinations of the incentive from the pie chart, it immediately emerges that in this case most of the incentive is allocated to covering expenses. The main reason is that, unlike the other cases, the low level of widespread self-consumption is reflected in a decline in the total incentive generated by the REC, so that expenses (assumed to be constant in the various simulations) acquire a certain weight.

It is also striking that in a situation of great imbalance between feed-in and withdrawal, the final incentive reserved for producers and prosumers is actually much lower than the share reserved for consumers. This is again attributable to the allocation percentages for the individual categories decided in advance.

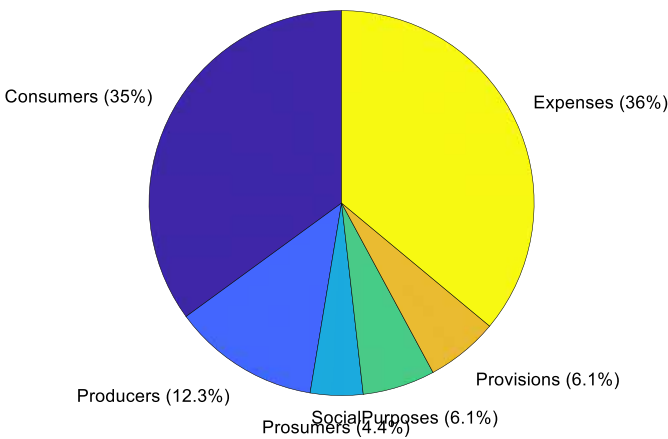


Figure 28: Incentive redistribution to categories - 4th simulation

Looking at the graph illustrating the incentive redistributed to individual users, the lack of a fixed quota is immediately striking, due to the minimal shared energy generated by the community. For this reason too, as can be seen from the graph, non-business consumers (in addition to having physiologically lower consumption than others) are significantly disadvantaged by the following situation.

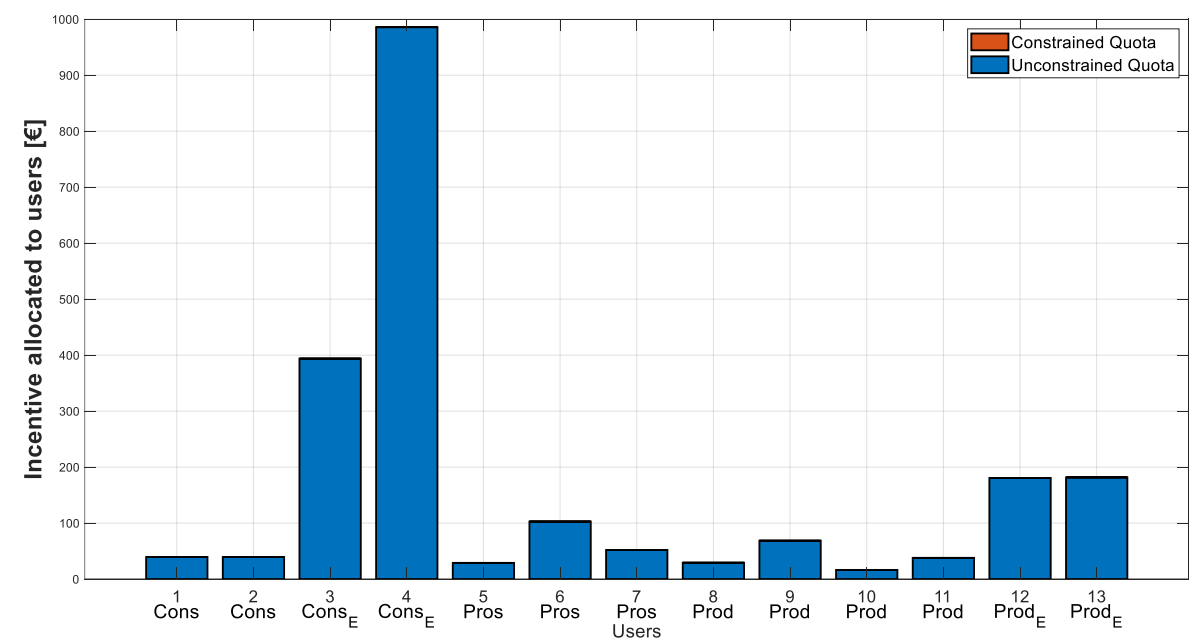


Figure 29: Incentive redistribution to users - 4th simulation

4.4.2. Comparison between total and unconstrained base allocations

A major decision point in the model is the choice of method for dividing the incentive released between plants that have benefited from capital grants and those that have not.

This step takes place after expenses have been covered, again starting from the gross amount of the incentive released.

The main doubt regarding this step is the choice of the allocation basis for the distribution described above. The first allocation basis consists of the total gross incentive, while the second consists of the gross unrestricted incentive.

To analyse the difference between the two methods, the incentive values distributed to each user were compared, first with one approach and then with the other.

This comparison was made for each of the simulations described above.

The final result of this comparison is a bar chart representing the incentive allocated to each user, using both approaches, in which the percentage difference between the unbound methodology and the “total” methodology is indicated for each user, taking the “total” share as a basis.

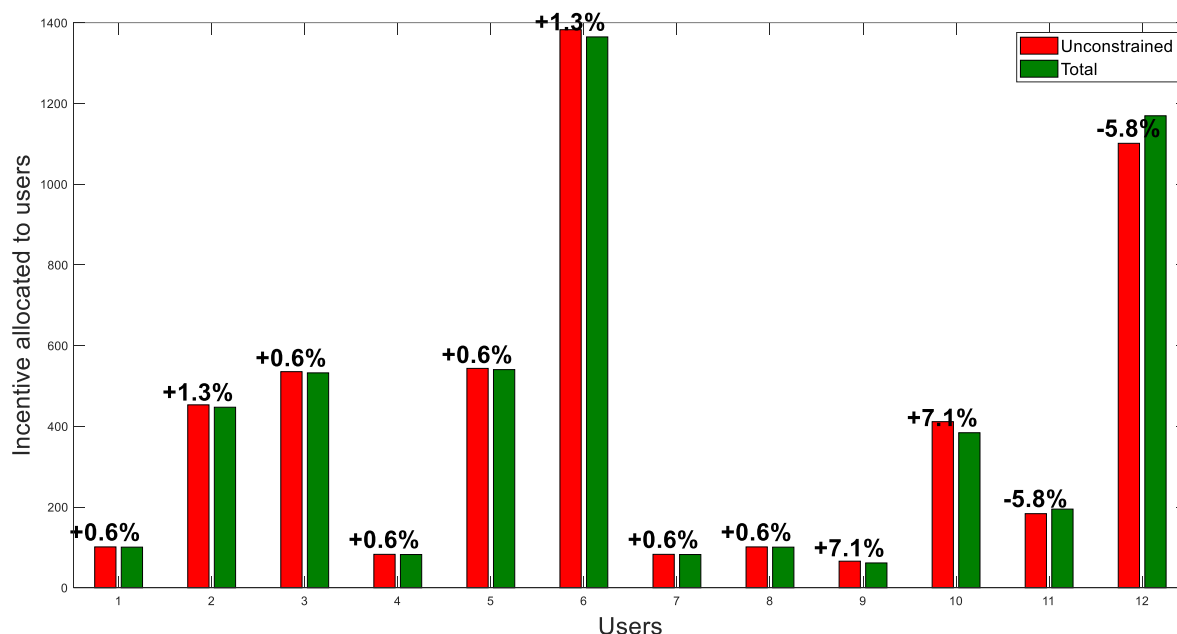


Figure 30: Comparison between total and unconstrained base allocations

What you see above is the diagram referring to the first simulation. The allocations for the two methodologies are actually quite similar, with a maximum difference between the two approaches of 7.1 percentage points.

The percentage trend of the difference between the two methods is similar for the second simulation, with a maximum difference limited to 1.2% in this case.

For the third case study, this difference increases slightly, given the significant amount of the restricted portion.

In the last case study, however, no difference is identified precisely because the unrestricted portion is zero, so the allocation basis is the same for both methods.

4.4.3. Allocation to consumers – Comparison between Proportional and Layered approach

An important part of the discussion of the results obtained from the simulations carried out using the model was the comparison between the two approaches proposed for the distribution of the incentive allocated to individual consumers, namely the 'Proportional' approach and the 'Layered' approach.

An initial observation arising from the analysis of the results provided by the four scenarios is that there is the following trend: consumers with lower withdrawals receive a larger share in the 'Layered' method than in the 'Proportional' method. The main reason for this is that, using the 'Layered' approach, they are often guaranteed the incentive on their total withdrawal.

At the same time, consumers with higher withdrawals receive a larger share in the 'Proportional' method, precisely because that method gives greater weight to the contribution to the total REC withdrawal and less weight to guaranteeing a certain share to the "weakest" consumers.

This behaviour is evident in all the four analyzed scenarios: smaller consumers are assigned a larger share with the 'Layered' approach, while larger consumers are assigned a larger share with the 'Proportional' approach.

The following figures illustrate the redistribution of the incentive to individual consumers for both approaches, for each simulation. At the same time, the total energy drawn by users is represented in orange (considering only the hours when shared energy is greater than zero), so as to distinguish between "large" and "small" consumers.

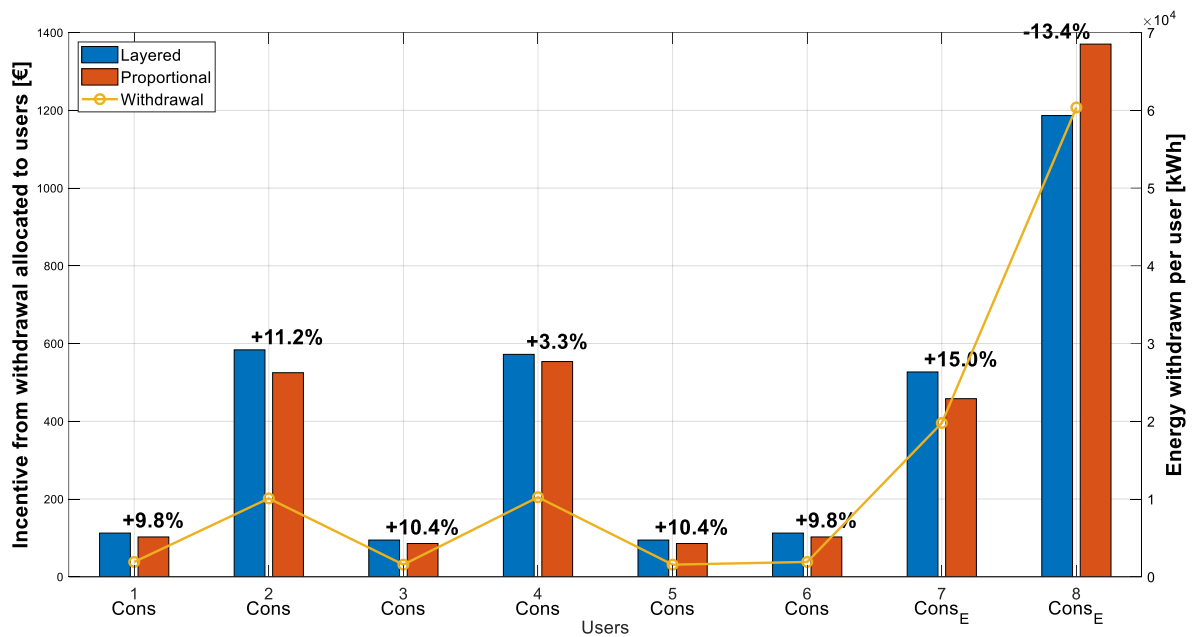


Figure 31: Incentive Redistribution Comparison - 1st simulation

Although the trend previously introduced can explain pretty well the behavior of small and big consumers, it alone does not provide a thorough explanation for users who have an intermediate absolute withdrawal value.

For this type of users, it is not a given that one approach will be more valuable than the other.

We have an initial demonstration of this in the second simulation (Figure 32):

By comparing the redistribution of consumer 5 and consumer 7, we realise that the logic proposed above does not seem to be correct. Consumer 7 has total annual withdrawals (considering only hours in which shared energy is greater than zero, approximately 29,274 kWh) greater than those of consumer 5 (considering only hours in which shared energy is greater than zero approximately 19,872 kWh). Despite this, consumer 7 benefits from the use of the 'Layered' approach, while consumer 5 benefits from the 'Proportional' approach.

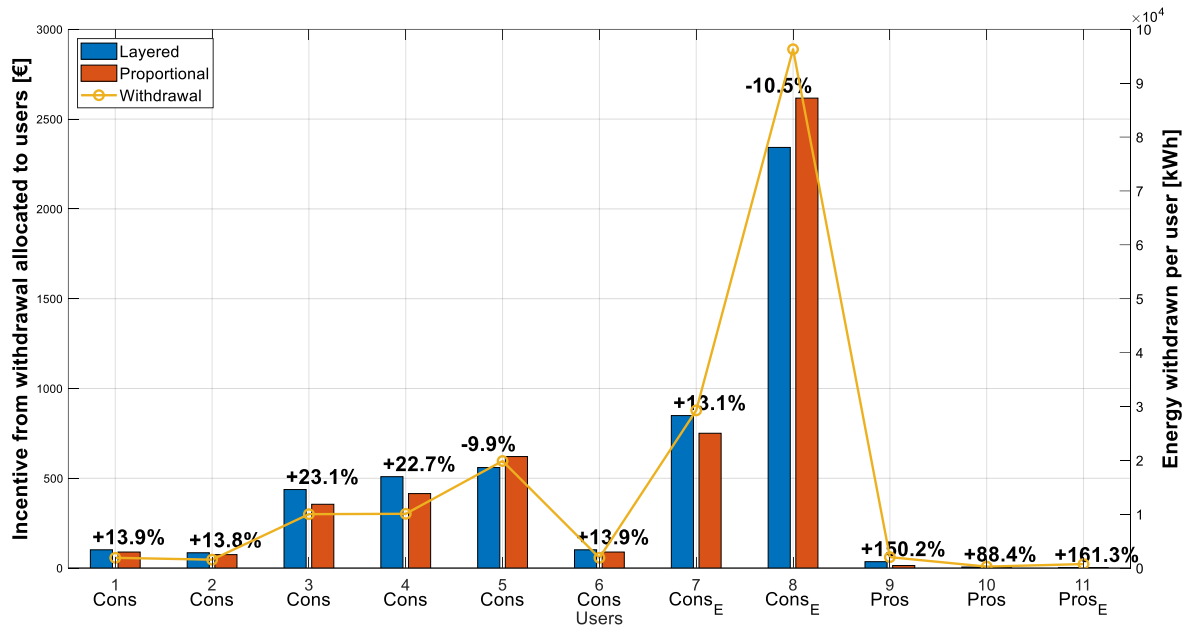


Figure 32: Incentive Redistribution comparison - 2nd simulation

In order to explain this type of pattern, the redistribution of the two consumers' consumption was analysed during a typical week throughout the year.

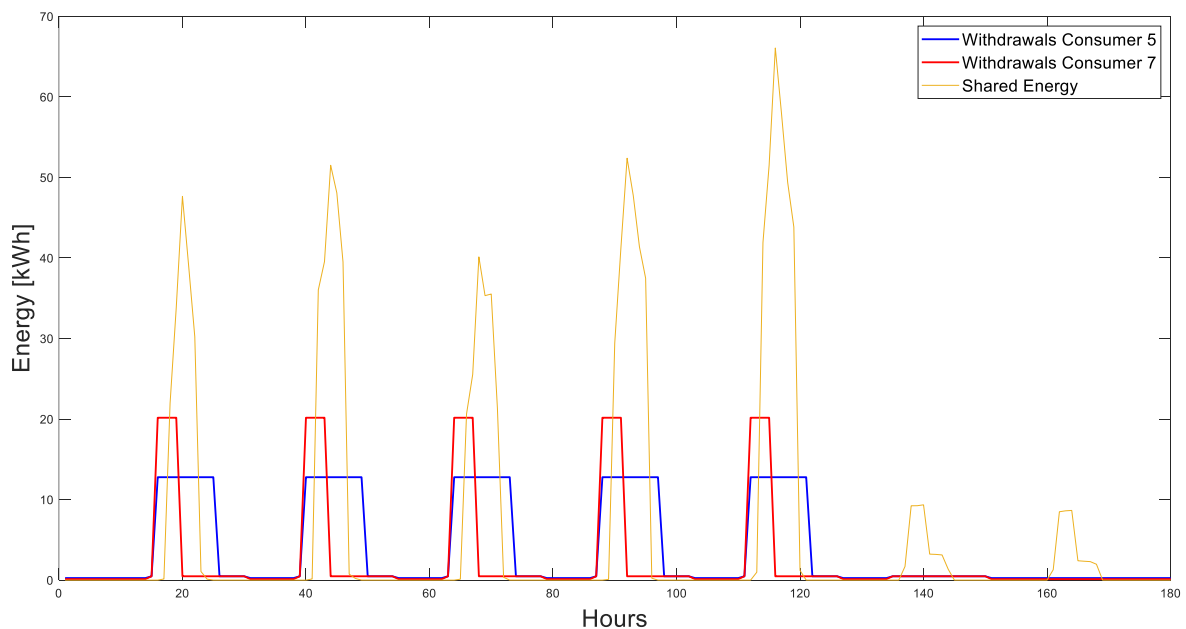


Figure 33: Withdrawal profiles for Cons5 and Cons7

The redistribution relating to Consumer 7, represented in red, shows clear and concentrated peaks: during certain time intervals, it absorbs much more energy than others. His withdrawals are more concentrated in a few hours, those of the morning,

in which photovoltaic production is low, and much higher in absolute terms than those of Consumer 5.

That relating to Consumer 5 (blue distribution), on the other hand, has a more regular and intermediate consumption profile: its consumption is less extreme, with more moderate peaks distributed over time and his hourly withdrawals are moderate and spread out over time.

Analysis of the two redistributions seems to reveal another clear trend:

- the *Proportional* model values the absolute quantity of consumption (rewarding those who withdraw more during moments of sharing);
- the *Layered* model values widespread and balanced participation, distributing the incentive more equitably among users with intermediate or constant profiles.

Reflecting on this, it was found that the ‘Layered’ method does not penalise consumption peaks as such, but redistributes shared energy according to a layered hourly logic.

As a result, if a user has very high consumption peaks at times when the availability of shared energy is limited, only part of their hourly consumption can be covered by the common layers. The rest of their demand remains uncovered and does not contribute to the allocation of the incentive.

Conversely, users with more moderate and consistent consumption profiles over time are able to participate in a greater number of hourly layers, receiving a higher average share of shared energy.

This is because, for every hour in which they participate, their consumption is neither minimal nor excessive: therefore, they always receive at least one layer of shared energy, and often more than one, because they remain “active” in all phases of hourly redistribution and benefit more often from the ‘Layered’ mechanism.

This effect explains why, when comparing methods, ‘Layered’ can reward “average” or regular consumers over those with more concentrated or intermittent consumption: multi-layer redistribution tends to favour the continuity and temporal compatibility of withdrawals rather than their absolute intensity.

Not yet satisfied with the empirical data analysed, it was decided to extend the research to the third simulation, in particular, analysing the trend of withdrawals related to Cons3. While the consumer behaviour was the previously identified association where high withdrawals correspond to a preference for a ‘Proportional’ approach among large consumers, we wanted to analyse the significant difference in incentive redistribution between the two approaches, which had not yet been observed.

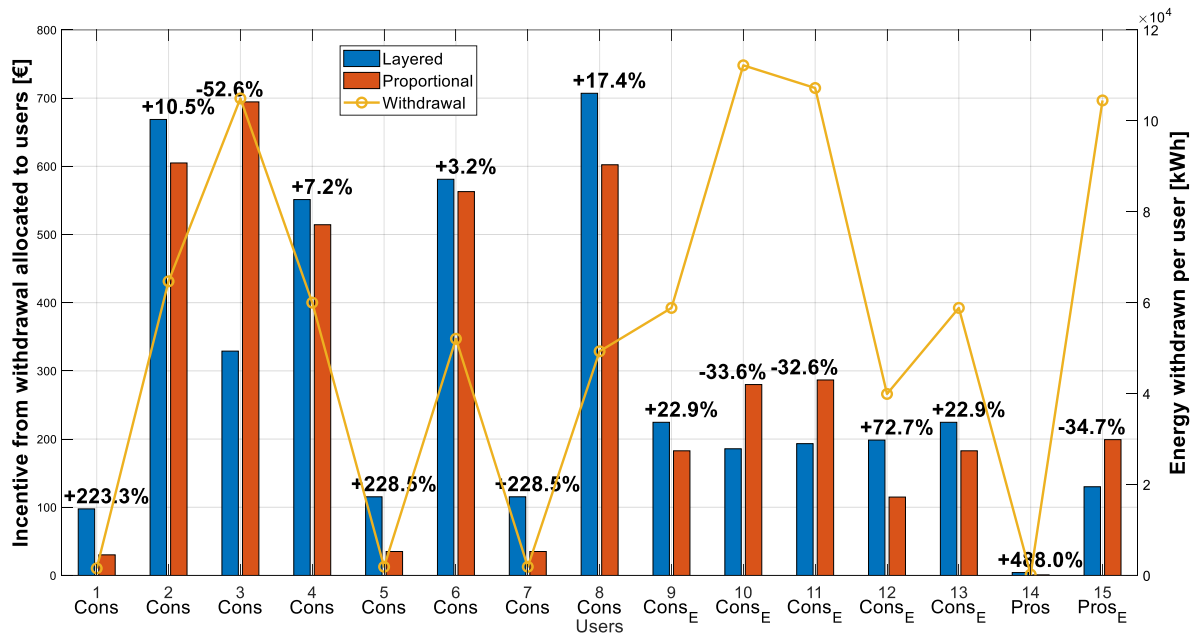


Figure 34: Incentive redistribution comparison – 3rd Simulation

To try to understand this behaviour and test the above hypotheses, the redistribution of the hourly consumption of the consumer was analysed, again considering a “typical week” during the year.

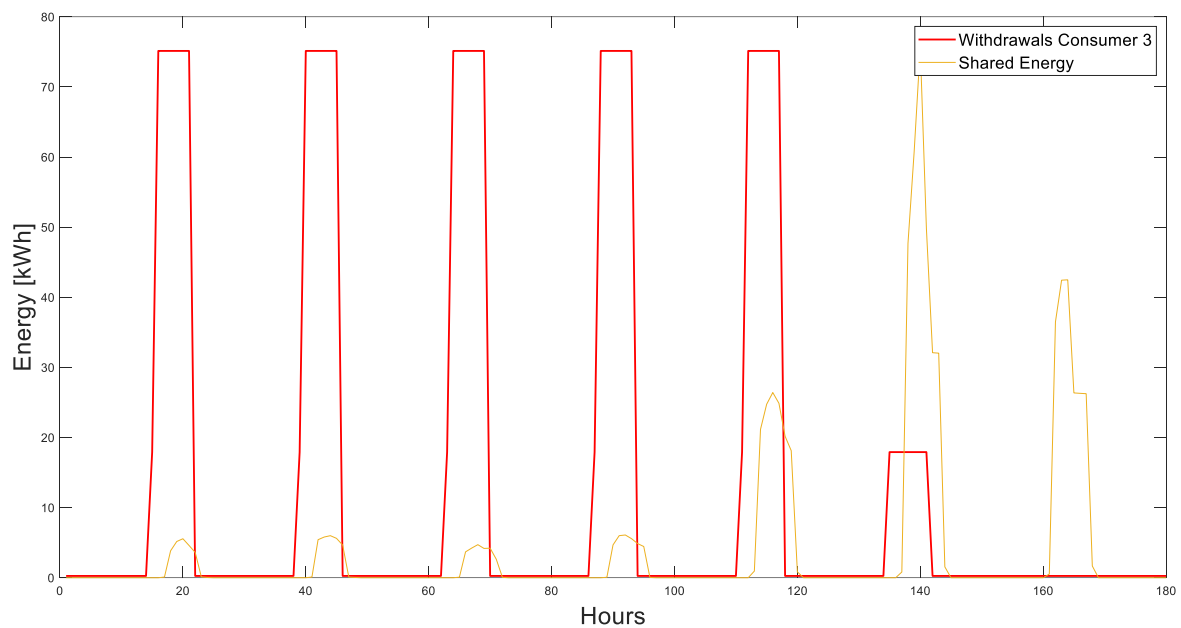


Figure 35: Withdrawal profile for Consumer 3

The observations made previously regarding the fact that the 'Layered' approach, with respect to the 'Proportional' one, tends to disadvantage consumers with high peaks of consumption have been confirmed.

Consumer 3 has highly consumption patterns that include numerous peaks, especially when shared energy is at medium-low level : for this reason, when adopting the 'Layered' approach, is that penalized, with respect to using the 'Proportional' one.

This is because it may happen that at times when consumption peaks occur, the availability of shared energy is exhausted before reaching maximum consumption levels.

Consequently, even if his consumption remains high, he does not receive an incentive that is fully proportional to his withdrawals, because part of his demand is not considered for shared energy allocation.

Withdrawal profile of Consumer 3 can be seen in Figure 35. Focusing on the moments when consumer 3 (red line) has peaks in consumption, the shared energy value, shown by the yellow line, is rather moderate.

If the 'Proportional' mechanism is used, most of that shared energy will be allocated to consumer 3, whereas if the 'Layered' mechanism is used, this would not happen, as the mechanism would first allocate the energy in such a way as to saturate users with lower consumption, and not necessarily recognise a greater allocation of shared energy to consumer 3.

A final analysis of the redistribution of the incentive in the last simulation using both methods was then done.

As shown in the figure, no particular trends other than those already mentioned above have been identified.

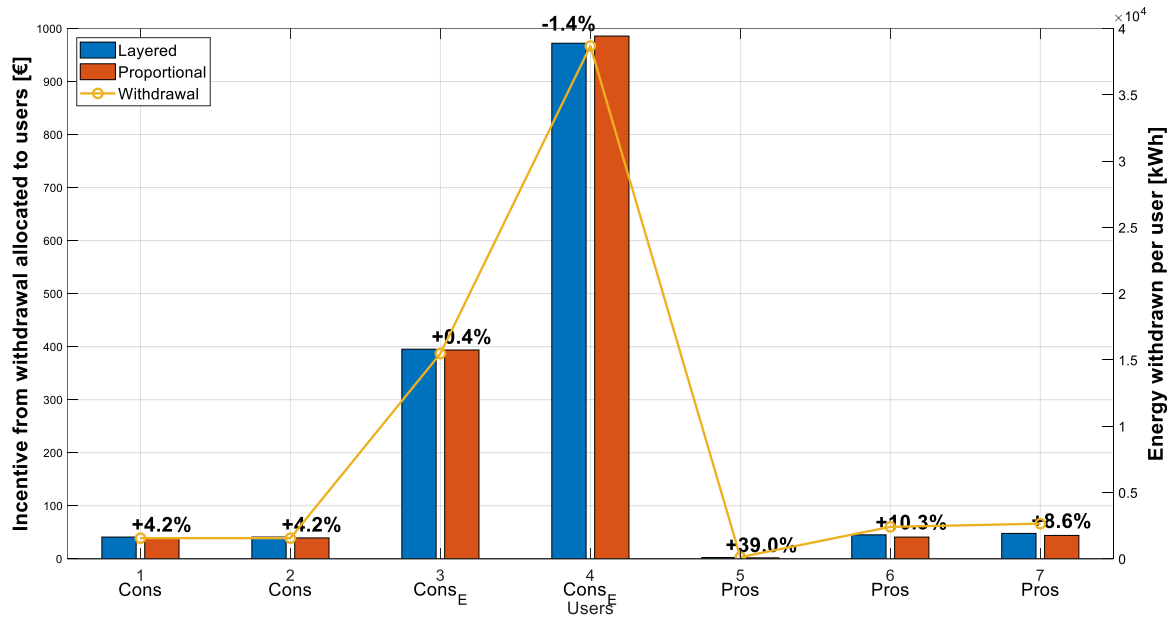


Figure 36: Incentive Redistribution comparison - 4th simulation

4.4.4. Gini Coefficient Comparison

A final phase of comparison between the two approaches aimed to examine the uniformity of distribution to consumers.

The Gini coefficient was used to quantify the degree of equity in the redistribution of incentives among users. Lower values of the coefficient indicate a more homogeneous and inclusive distribution, while higher values denote a greater concentration of incentives on a few individuals.

This coefficient is particularly suitable for describing how concentrated a resource is among a few individuals or how evenly and usefully it is distributed when assessing equity, uniformity or distributive justice, not just numerical variability.

The Gini coefficient was calculated for each simulation and for both approaches. The results are summarised below:

Table 6: Gini coefficient values for each approach in the various simulations

SIMULATION	GINI PROPORTIONAL	- GINI - LAYERED	Δ GINI (PROP – LAY)
1	0.6622	0.6350	0.0272
2	0.7073	0.6738	0.0335
3	0.4888	0.4417	0.0471
4	0.8265	0.8225	0.0040
AVERAGE	0.6712	0.6432	0.0280

The results show that, by averaging the Gini coefficients across the various simulations, the approach that redistributes the incentive most evenly to consumers is the 'Layered' approach.

In specific cases, such as simulation four, with a low level of self-consumption, the difference between the two approaches in terms of the uniformity of redistribution is almost negligible.

5 Validation and applicability of the model

5.1. Applicability of the model - feedback from interviews

Objective and Discussions

Once the model had been finalised and the simulations analysed, interviews were conducted with individuals involved in the REC design, operation and financing to gather their opinions on the incentive redistribution methodology and the accuracy of the model, and to evaluate any upgrade.

The following interviews were conducted on two separate occasions.

The first was conducted during the REC Observatory led by ENEA, focusing mainly on the redistribution of economic benefits within RECs. The second was a more focused interview with Paolo Benfenati, president of REC WeVèz, an energy community based in Bologna, and Paola Bellotti, director of Sustainability and Development at Coopfond S.p.A..

Both occasions gave rise to interesting reflections on the structure of the proposed model, but also on some very interesting issues related to the purposes for which the incentive should be used.

Below is the feedback and information gathered, which is important for validating the proposed model and completing my research on the purposes of incentive redistribution.

General structure and purpose of a redistribution methodology

When discussing the main characteristics that an allocation method should have, simplicity and applicability emerged as particularly important, in addition to fairness.

The basic idea behind the model was to create a tool for RECs that was as “generic” as possible and that also incorporated the analysis of cases/scenarios that are not usually common in reality. Obviously, all the choices made in compiling the model sought to ensure the highest possible level of fairness.

Regarding the structure of the proposed model, feedback was received from the President of the REC WeVèz, Paolo Benfenati.

He insisted on the importance and correctness of dividing the incentive into various categories or clusters, and then using specific redistribution strategies within the same cluster.

With regard to the structure of the redistribution methodology, positive feedback was received concerning the initial distinction between the restricted and unrestricted portions of the incentive. The model would solve two major problems that have emerged: firstly, it would simplify the calculation of the restricted and unrestricted quotas, which many REC managers describe as “ambiguous”. Secondly, the division of the two separate quotas would ensure that RECs comply with the restrictions imposed by the Italian regulatory framework, channelling the incentive towards one destination rather than another.

Advance payment adjustment mechanism

During the aforementioned meetings, doubts arose regarding the inclusion of the advance payment-adjustment mechanism model proposed by the GSE within the operating rules.

The model was set up with a “retrospective” logic, meaning that it is basically suitable for redistributing the incentive only “on a final balance basis”.

This choice was made both for reasons of simplicity and because the initial objective was to manage the redistribution of the economic benefits actually accrued by the REC during the year.

A further division taking into account the advance payment-adjustment mechanism would have made everything more complicated and would not have generated any surplus for the initially set objective.

For this reason, the model will be particularly useful to RECs once they have received their annual adjustment from the GSE, together with the withdrawal, feed-in and shared energy data provided by the GSE.

Allocation percentages to categories

Another much-debated topic during the meetings was the percentage allocation reserved for each macro-category in a redistribution methodology.

What emerged from the various discussions was that these percentages are highly dependent on the individual REC: on its legal form (and therefore on the constraints imposed by having adopted a particular legal form rather than another), on the size of the REC, but above all on the purposes for which the REC was conceived and then created.

Although this had already emerged from the analysis of the regulations of the RECs currently existing in Italy, two diametrically opposed views emerged during the interviews. The first was provided by a manager of the REC “Le Vele”. This REC, established as an association, has the stated aim of supporting social projects and

vulnerable people. For this reason, the total incentive obtained is allocated to these purposes, even at the expense of an increased payback for the prosumer (public administration).

The second view emerged from Paola Bellotti, who is primarily involved in financing RECs in the form of cooperatives. In their case, a large part of the proceeds obtained from the state is destined, also due to imposed constraints, to remain in the cooperative as a provision fund.

For this reason, those who discussed and debated this issue agreed to leave the choice of allocation percentages to the individual RECs, as a variable to be entered by the external user.

As mentioned above, intermediate values emerged from the analysis of Italian regulations carried out in Chapter 3 were used in the simulations.

Expenses Coverage

The debate then moved on to the topic of ‘expense coverage’, in particular what had emerged from the analysis of REC regulations. First of all, it was confirmed that expenses can only be deducted from the non-excess incentive quota. After that, the two methods most commonly used by RECs to cover costs were discussed. The first consists of allocating a fixed percentage to them, while the second redistributes the uncommitted incentive “net of REC expenses”.

Although it was agreed that the best alternative, as proposed by the model, was to redistribute “net of expenses”, Paolo Benfenati made an important point about the advantage of allocating a percentage of benefits to cover costs.

The great advantage of allocating a fixed share to costs, typically left generous enough to cover even exceptional cases, is to protect against possible opportunistic behaviour by REC managers, which could “allocate” themselves higher costs than they actually deserve.

Another point of discussion was the use of provisions to cover costs, but this topic will be explored in more detail in the section on provisions.

Distinction between treatment of plants benefiting or not from subsidies

The distinction between producers who had received capital grants and those who had not was a particularly debated topic.

Although the participants in the meeting recognised the importance of treating the two categories differently, both because of the different economic support received by the two and because one of them contributed to the reduction of the incentive and the other did not, the adoption of this distinction within a REC regulation was not well received. For Paolo Benfenati, the main reason why RECs, including his own, do not always distinguish between the two categories is because it is too complex. In an Italian regulatory landscape that is already particularly complicated, he argues that a further

subdivision between the two categories is particularly difficult to manage and to communicate to the members of the REC itself.

Nevertheless, as mentioned above, the different roles of the two categories within the REC have been recognised, and therefore the use of the distinction between the two categories within the model we proposed was ultimately approved.

Redistribution of Net Non-constrained Incentive

Provisions

With regard to the category of provisions, a clarification emerged from the general debate. This clarification refers in particular to RECs in the form of cooperatives, because by law they must set aside part of their turnover.

There are two types of provisions: a first “legal” provision, corresponding to 3% of revenues, and an “indivisible reserve”, equal to 30% of revenues.

It is important to note that the first type of provision must be made before deducting REC costs, while the second is made after covering expenses, starting from net profit. For this reason, while the first should be placed within the model as an item preceding cost coverage, the second is correctly attributable to the “Provisions” category, to which the incentive is directed.

Social Purposes

The issue of social objectives generated particular interest in the debate.

With regard to the inclusion of a quota for social purposes within the model, for the same reason discussed above regarding the REC's independence in choosing how much incentive to allocate to each category, the decision to leave a percentage of the excess incentive for social purposes was approved.

As regards the percentage share in relation to the uncommitted incentive, some concerns were raised, as it is very rare for RECs to allocate part of the non-excess premium tariff to social purposes in addition to the portion relating to the committed share.

At the same time, however, it was recognised that, in order to maintain a certain level of generality and broaden the applicability and usefulness of the model, it was necessary to at least provide this opportunity to RECs.

Another topic of debate was the social activities that a REC can implement. The initial idea behind the model was to try to understand, for each quota allocated for social purposes after redistribution, the possible activities/proposals to be financed. In this regard, the question was asked how this quota is currently used in RECs and whether it would be possible to implement something similar in the model. According to the experts, it is very difficult to define a list of activities that can be categorised as Social

Purposes. This varies greatly, both in terms of the purpose of the REC itself and the characteristics of the territory and environment in which the REC is located.

The REC WeVèz, for example, gives vouchers to users in need, to be spent exclusively in the territory where the REC is located. In doing so, it contributes to fuelling the circularity of the REC initiative, further developing companies located within the REC territory and reviving the spirit of initiative of RED II. The REC “Le Vele”, on the other hand, has set up a special fund to combat energy poverty in collaboration with social workers.

Another example is that of a REC where one of the users is the company that manages the town's sports field. In that case, the social purpose quotas are allocated to the development and growth of facilities and infrastructure to be made available to the local community.

The decision on what activities to carry out with the portion allocated to social purposes is seen much more as a ‘guidance’ mechanism than a process that can be engineered.

During the interview, I then asked for information about any indicators used to identify vulnerable users or energy poverty conditions. The only information I found was provided by Violetta Scipinotti, in reference to REC Le Vele.

The REC itself does not use any indicators to determine who can be considered a vulnerable user and who cannot, but relies on the support of social services. The latter refer to indicators such as ISEE (Indicatore della Situazione Economica Equivalente) and others that were not mentioned.

A final interesting point emerged from Paola Bellotti's speech, referring to the use of a portion of funds for social purposes. In her view, in many situations the development of a REC is already seen as a social and local development initiative, so within redistribution strategies, RECs tend to give less emphasis to this aim and purpose.

Allocation logic for individual user

The last part of the model discussed in my interviews refers to the logic of redistribution to individual users, whether they are consumers, producers or prosumers.

In this regard, respondents were asked how they believed the incentive should best be allocated to individual consumers/producers, and in particular, what parameters they considered appropriate to include in the model to approximate the contribution made to the REC by each user.

All interview participants agreed that the incentive should be allocated to individual users in proportion to each user's contribution to shared energy.

Specifically for the “producers” category, they all agreed on the allocation to individual producers proposed by the model, which rewarded both the contribution

to the feed-in by the individual plant and how useful this feed-in was in increasing the final incentive quota.

As regards the consumer category, the topic sparked some debate. Some members of the meeting argued that this was unnecessary, as the GSE's indications already clearly showed the amount of shared energy consumed by each consumer.

However, analysis of the report provided by the GSE clearly showed that the GSE only reports the energy withdrawn by each user in that time slot, without providing any indication of how the shared energy can actually be distributed at times when withdrawal exceeds input among consumers.

Following this clarification, the two approaches of the model were discussed and both were approved as they take into account the contribution to shared energy by each consumer. No particular indications emerged regarding a preference for one over the other.

Distinction between natural persons and legal persons

Part of the discussion with respondents focused on an issue that the model did not address. This issue concerns the distinction between the treatment of natural persons and legal entities within the redistribution strategy. Paolo Benfenati, manager of the REC 'WeVèz', stated that it is essential to distinguish between the two categories when allocating, and then redistributing by clusters that separate between natural persons and legal entities.

The main explanation lies in the fact that very often legal persons within the REC are characterised by much higher inputs and withdrawals than natural persons. Distributing the incentive without this distinction would result in a sort of "cannibalisation effect", in which large consumers (and producers) would monopolise the redistribution of the incentive, leaving little or nothing for natural persons in the REC.

However, this effect can also occur in the opposite direction. As Paola Bellotti points out, there are cases in which the REC is unbalanced from the point of view of natural persons. In her view, if the number of natural persons within a REC increases too much, it would be necessary to introduce a mechanism such as the one proposed by Paolo, which in this case would act as a "defence" for the categories of producer/consumer enterprises.

From this discussion that arose during the interview and from this debate, it was decided to evaluate this initiative in concrete terms, and this will be shown in the next section on model refinement.

5.2. Model refinement

The interviews and discussions provided valuable feedback on the strengths and weaknesses of the model. Among all the feedback, the issue that seemed to require further investigation was the lack of a clear distinction in the model between natural persons and legal entities.

For this reason, it was decided to analyse the simulations carried out to try to understand whether there was indeed a tendency within RECs towards the so-called “cannibalisation effect”.

To do this, the percentage share of the incentive allocated to individuals was calculated for each simulation. This was done starting from the incentive allocated to consumers (linked to energy withdrawal), the incentive allocated to producers (linked to energy injection) and the total incentive.

The approach used was the ‘Proportional one’.

The results of the analysis are shown in the following figure:

Table 7: Incentive quotas allocated to individuals for each simulation

Simulation	% of Incentive from withdrawal allocated to Individuals	% of Incentive from injection allocated to Individuals	% of total Incentive allocated to Individuals
1	8.84%	14.15%	6.66%
2	5.10%	10.39%	5.19%
3	2.34%	4.58%	1.86%
4	5.21%	11.88%	3.68%
Average	5.36%	10.25%	4.35%

The figures provided in the table show that a particularly small percentage of incentives is allocated to individuals, confirming the need to divide the allocation into clusters that also consider the distinction between individuals and non-individuals, thereby avoiding the so-called “cannibalisation effect”.

Following careful review of these analyses, it was therefore decided to implement a further distinction in the model between natural and non-natural consumers, introducing an additional distinction between legal entities that are businesses and those that are not.

The final structure of the model is as follows:

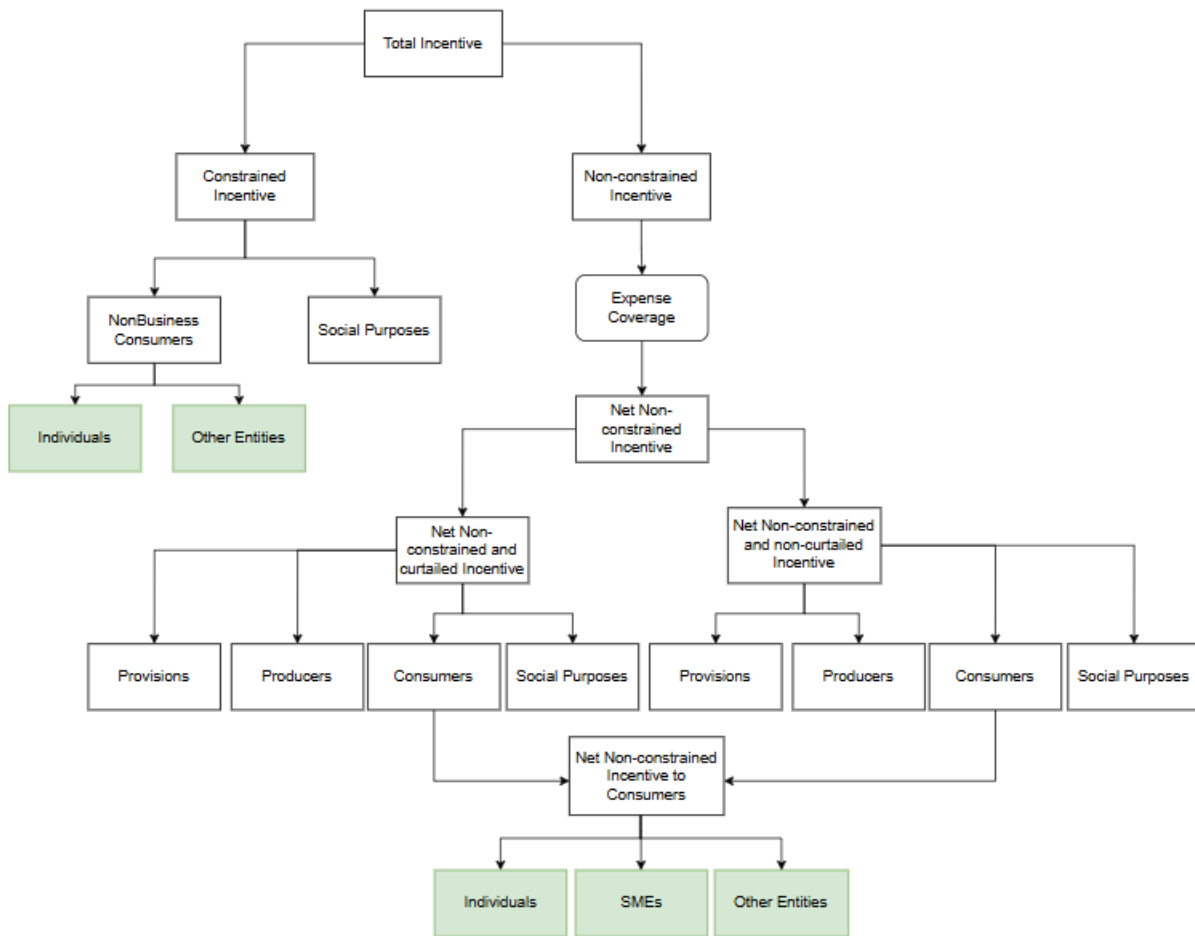


Figure 37: Final structure of the model

By modifying the model in this way, it was necessary to introduce new input parameters into the model.

The first is the percentage allocation of the restricted incentive due to non-business consumers in two categories: individuals and other entities. In this case, the “small and medium-sized enterprises” cluster was not considered, precisely because of the restriction imposed by the legislation.

For the constrained incentive allocated to Non Business Consumers:

- Allocation percentage to individuals: 50%
- Allocation percentage to other entities: 50%

For the Net Non-constrained Incentive allocated to consumers:

- Allocation percentage to Individuals: 20%
- Allocation percentage to SMEs: 50%
- Allocation percentage to other entities: 30%

After modifying the model as described above, the same simulations analysed previously were run to understand the effectiveness of the change adopted to protect individuals within the REC.

The results shown in the table indicate a 390% increase in the share allocated to individuals, based on the total incentive allocated to the withdrawal category, and an 195% increase based on the total incentive.

Table 8: Incentive quotas allocated to Individuals - Enhanced Model

Simulation	% of Incentive from withdrawal allocated to Individuals	% of total Incentive allocated to Individuals
1	27.52%	14.23%
2	24.07%	13.14%
3	33.85%	14.77%
4	20%	9.17%
Average	26.36%	12.83%

These results show a clear increase in the quotas allocated to individuals within the RECs. It is important to emphasise that these results were obtained through allocation using the percentages described above, so a valid future analysis could seek to identify an identifying value for the allocations just described.

5.3. Discussion of the model and link with literature gap

In this part of the paper will be analysed how the proposed model has addressed the gap that emerged from the literature review, and how it has done so by best integrating the specificities of the Italian regulatory landscape.

It will be analysed how the model integrates the three main aspects to be taken into consideration in a redistribution strategy that can be considered comprehensive, namely the technical, economic and social aspects and at the same time fully comply with the restrictions imposed by the Italian regulatory framework.

Before going into detail on these aspects, it would simply be emphasised that a redistribution strategy based on “clusters” has been established for the structure of the model, i.e. a tree structure starting from the total incentive and ending with the incentive allocated to each category. Within each of these clusters, the incentive would then be further allocated to individual users belonging to the specific cluster.

The basic idea of the model is based on the ‘benefits redistribution tree’ adopted by Trevisan in ‘Economic benefits redistribution methodology for Renewable Energy Communities’ [20], but it has been significantly modified to reflect the guidelines of the Italian regulatory framework and to add some new features.

5.3.1. Technical Dimension

As regards the technical aspect covered by the model, the basis for the allocation was the levels of energy shared by each plant within the REC. These were used as the starting point for identifying the two incentive quotas, restricted and unrestricted, for each of the two categories of plants (those benefiting from a capital grant and those not benefiting from one).

This distinction between the two quotas, specific to the Italian context, is a novelty compared to what is proposed in the literature, probably also due to the strong focus on the Italian context that characterises the model. Despite this, no distinction between the two quotas, surplus and non-surplus, as proposed in the model, had yet been identified in the literature.

Analysing the allocation at individual user level in greater detail, the model basically used parameters that had already been analysed in the literature. The three main parameters are:

- Hourly energy fed in by each producer [kWh]
- Hourly energy withdrawn by each consumer [kWh]
- Hourly shared energy[kWh]

This part of the model bears some similarities to the method proposed by Fioriti [6] and Trevisan [13], [20]. They also analyse all these variables in their allocation proposal, i.e. the contribution to shared energy by each user. However, unlike our model, Fioriti's paper does not specify the underlying mechanism for calculating the shared energy associated with each producer and consumer.

In Trevisan, the shared energy is associated proportionally to the energy injection and withdrawal of each user. However, the time period taken into account for this allocation is not specified.

The approach used by our model to allocate the incentive to individual producers is innovative.

It takes into account the different value of the inputs from each producer. In other words, the model seeks to reward as fairly as possible the producer who actually contributes to the shared energy of the REC and, therefore, to the total incentive.

Using the formula explained in the previous chapter, a higher incentive is therefore guaranteed to producers who produce a lot at times when the total shared energy is

greater. In this way, a distinction is made between inputs that actually contribute to an increase in the REC incentive and those that make a smaller contribution.

By focusing instead on the redistribution of the incentive associated with energy withdrawal and allocated to each consumer, the model offers two possible approaches, both with the aim of allocating shared hourly energy to individual consumers, based on each consumer's consumption during that hour.

The first allocates shared energy proportionally to the energy consumed by each individual consumer, relative to the total energy consumed by users.

The second, as already explained in the previous chapter, distributes the shared energy progressively from users with lower withdrawals to users with higher withdrawals. This second algorithm is taken from academic literature and proposed by Minuto [3] and has been adapted to MatLab code for implementation in the proposed model.

Since the model has a decision tree structure, whereby hypothetically each user can use it with their preferred settings and purposes, it was decided to retain both of these distribution options.

5.3.2. Economic Dimension

With regard to the economic dimension, the model first includes REC expenses as a parameter to be entered by the user. As indicated by national legislation, these are deducted from the non-excessive incentive quota. In particular, the model covers expenses starting from the uncommitted incentive, and then continues with redistribution. It is important to emphasise this distribution because in the literature, for example in Trevisan [20], the share relating to expenses is often deducted from the total incentive, and no mention is made of the distinction between the restricted and unrestricted portions.

Analysing the comparison with the literature, it is immediately apparent that within our model, the total incentive is divided into numerous quotas. The first subdivision is the one discussed above regarding the difference between Restricted Incentives and Unrestricted Incentives. It was considered essential to include this distinction within the model because the two portions, due to regulatory constraints, must be allocated to two different purposes. This subdivision is a novelty in the literature, which tends not to mention the two distinct portions, but treats the incentive indiscriminately.

Another subdivision that is new to the academic framework is the distinction between incentives from plants that have had access to capital grants and plants that have not had access to any grants. The financial contribution received from the state or other European Community funds then becomes a fundamental parameter for the redistribution strategy, as it allows for differentiation between the treatment of plants that have received non-repayable grants and plants that have not received any grants.

This distinction was deemed necessary as it was considered fair to differentiate the allocation percentage for plants that, by receiving a grant, contributed to a reduction in the incentive tariff, thereby negatively impacting the total incentive amount to be redistributed to REC members.

5.3.3. Social Dimension

With regard to the social dimension, in seeking a model that would include the needs and objectives of every possible REC, it was decided to leave the percentage allocation for social purposes to the discretion of the external user.

There was also much discussion about the possibility of including indicators related to energy poverty and the social vulnerability of households. Although this is a useful tool for subdivision, it was considered too complicated to access information that is often linked to individual households participating in the REC. For privacy reasons, it would be complicated to include indicators (such as the ISEE) in the model, which could only be used after direct consultation with users and with the support of third-party associations such as social services.

One parameter associated to the uniformity of redistribution, particularly that relating to consumers, was the Gini coefficient. The Gini can be used as a quantitative proxy for the fairness of the model as it gives us an indication of how excessively unbalanced the incentive system is or how “fair” the system actually is. In particular, within the model, it is used to analyse which of the two mechanisms, “Proportional” and “Layered”, redistributes the incentive more evenly among users.

Following discussions and interviews, it was also deemed appropriate to include a distinction between individual consumers and corporate consumers, to prevent consumers with higher withdrawals from receiving an excessive share of the incentive, which would have a “cannibalisation effect” on users with lower withdrawals (typically residential households).

The inclusion of a specific cluster in the model to support individual consumers, including users and families in economic hardship, is a novelty in the literary landscape.

Through this strategy, a certain incentive quota is guaranteed to the most vulnerable users of the REC, and the so-called “cannibalization effect” by companies or entities with higher withdrawals, toward residential users, is avoided.

6 Conclusions

In this final chapter, we will analyse the implications of implementing the model, both in a theoretical context and in a practical context. We will analyse its limitations and focus on what future developments may be, also looking at the evolution of the European context.

6.1. Theoretical implications

As regards the contribution made by the model to existing academic literature, as already discussed in Chapter 5, the model sought to bring together three dimensions considered fundamental by many of the papers already analysed: the technical, economic and social dimensions. The further step introduced by the model was to try to consider all three of these aspects, whereas in most of the papers analysed, these aspects were dealt with individually, or at least not in an integrated manner.

Another characteristic feature of the model, which had not yet emerged in the literature, was its high degree of specificity with regard to the Italian context. According to the author's analysis, existing literature lacks in dealing with the development in such detail of a redistribution strategy that integrates perfectly with the constraints and rules imposed by the Italian regulatory framework.

Another distinctive feature of the model that distinguish itself is the generalizability of application: it has been designed to adapt to various types of RECs in order to meet their various aims and objectives.

6.2. Implications for operative RECs

The proposed model can serve as an operational reference point for Italian Renewable Energy Community (REC) managers in dealing with a complex and still evolving issue such as the redistribution of economic incentives. In a context characterised by significant differences in interpretation and a regulatory confusion, the model provides a solid basis from which to systematically address the calculation and management of the constraints imposed by the national regulatory framework.

The structure of the model allows each manager to customise the allocation of the incentive according to the specific characteristics and objectives of their energy community, while ensuring compliance with current regulatory constraints. This makes it possible to maintain a balance between management flexibility and regulatory compliance, promoting the autonomy of individual REC and their ability to adapt to different territorial and social contexts.

From an operational point of view, the proposed model is a valuable management support tool, useful not only for managing regulatory obligations, but also for monitoring and optimising the mechanisms for allocating economic benefits among the different categories of participants. This approach helps to improve the transparency, fairness and efficiency of the redistribution system, which are essential elements for consolidating member confidence and ensuring the long-term sustainability of the CER.

Furthermore, the adoption of the proposed model can promote a higher level of accountability and internal reporting, improving the traceability of decisions and communication with stakeholders. In this sense, the model is not only a technical tool, but also a means of promoting more participatory and transparent governance practices, in line with the founding principles of energy communities.

6.3. Limitations

The proposed model has some limitations that should be highlighted to correctly interpret the results obtained. Firstly, having been developed specifically to comply with Italian regulatory constraints, its applicability to the European context is limited: the model reflects regulatory peculiarities that may not be immediately transferable to other European Union countries. Consequently, its use in different contexts would require preliminary adaptation to specific legislative provisions and different incentive mechanisms at local level.

More specifically, the application of this model would perhaps be more feasible in European countries that attribute a tariff remuneration to shared collective energy, as is currently the case in Italy, Belgium, Greece, Germany, Portugal and Spain [3].

Secondly, in order to maintain a high degree of flexibility and generality of use, the model requires a series of input parameters to be provided by the user. Although this approach offers a high degree of flexibility and adaptability to different community configurations, it requires managers to have a thorough understanding of the internal logic of the model and the principles governing the redistribution of benefits. Consequently, it is important to accompany the adoption of the tool with adequate training and support, so as to ensure its informed use in line with the objectives of the energy community.

Finally, although some of the simulations carried out have attempted to reproduce the performance of a real Renewable Energy Community, the model has not yet been tested on concrete cases. Applying it to a real case study would be an essential step in verifying the model's ability to respond to empirical inputs and assessing its actual effectiveness in managing economic flows and redistributing benefits. Such applications to real cases therefore represents an important direction for future developments and improvements.

6.4. Proposal of future research

As the final part of the conclusions, possible areas for improvement of the model and future developments are suggested.

First of all, as we mentioned earlier, applying the model to real case studies could demonstrate strengths and weaknesses of the proposal put forward in this thesis.

This possible next step could provide a more in-depth understanding of the calibration of the input parameters to be used, test the robustness of the model in the presence of real configurations, and perhaps identify new variables or indicators useful for improving the accuracy of redistribution. Finally, contact with real cases would provide direct feedback from managers and community members, which would be useful for making the model more responsive to practical needs, thereby increasing its social acceptability.

Another aspect of the model that could be explored further is the development of social objectives. Following privacy concerns, the issue of how to include support for vulnerable users and those in energy poverty has been partially sidelined. The very concepts and definitions of vulnerability and energy poverty still seem rather inconsistent and varied within the academic world and beyond, which has slowed down the possible development of the model in this direction. At the same time, however, further exploration of these aspects, particularly how to include steps in the model that take into account vulnerable users or those in energy poverty, could represent an area for future research.

Future additions to the model will be necessary due to the evolution of European legislation. On 8 October 2025, the Council of Ministers gave preliminary approval to the draft decree implementing the 2023 European Renewable Energy Directive, RED III.

RED III seeks to streamline procedures for the integration of renewables into the European energy landscape and provides clear guidelines for the development of renewable energy communities.

An example of this is the reduction in the time required for permit-granting procedures, with Member States required to ensure that the permit-granting procedure for the installation of solar energy equipment with a capacity of 100 kW or less shall not exceed one month, after which, without a response, the permit is guaranteed. Another example is the proposal to introduce simplified registration processes and reduced registration fees.

Another point on which RED III focuses for the development of RECs, and which will need to be taken into account within the model, is the participation of RECs in the battery and electric vehicle markets, through flexibility services such as demand response and storage mechanisms.